

Coarse Revenue Guarantee in First Price Auction and Beyond*

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Abstract

I study the coarse revenue guarantee (CRG) of the single-unit first-price auction with common values. The CRG is the minimum expected revenue over all information structures and all Bayes coarse correlated equilibria (BCCE), the equilibrium notion to which no-regret learning leads. The worst case is identical play, in which all bidders submit the same bid, and this reduction yields a closed-form characterization. The coarse guarantee is positive and increases with the number of bidders N . It lies strictly below the interim guarantee of [Bergemann, Brooks and Morris \(2017\)](#) at every finite N , and the gap between the two closes at rate $\log N/N$. I extend the analysis to reserve prices, symmetric and asymmetric priors, other standard auctions, and an unknown value distribution.

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1 Introduction

The robustness of mechanisms to the information structure is a central question in mechanism design. A seller who knows the distribution of the good’s value but not the buyers’ information structure may want a lower bound on revenue that holds across every environment consistent with that prior. [Bergemann, Brooks and Morris \(2017\)](#) provide such a bound for the first-price auction. They characterize the **revenue guarantee** (RG), defined as the minimum expected revenue across all information structures and all Bayes correlated equilibria (BCE), and show that it is positive, even as the number of buyers grows large. The BCE framework assumes fully Bayesian rationality. In a BCE, a bidder who is told to submit a_i after observing signal s_i must be unwilling to change that bid *at that signal*. Obedience holds signal by signal, which requires each bidder to process their private information and best-respond conditional on it. This is the natural requirement when bidders are sophisticated Bayesian agents.

But it may not be the appropriate requirement when bidders are algorithms. In modern electronic marketplaces, and increasingly in advertising exchanges and high-frequency trading venues, bids are submitted by automated agents. These agents adjust their bids over time by simple learning rules rather than by solving a Bayesian inference problem. The standard performance criterion for such rules is *no regret*: the algorithm guarantees that, in hindsight, no single fixed action played throughout the history would have earned more on average. That is an unconditional guarantee. It says nothing about whether the algorithm’s behavior would have been optimal in any particular state, and a no-regret algorithm can have large regret conditional on individual states without ever violating its guarantee.¹

The solution concept that addresses this weaker notion of rationality is Bayes coarse correlated equilibrium (BCCE), in which each player must be unwilling to switch to a single fixed action played at every signal. [Hartline, Syrgkanis and Tardos \(2015\)](#) show that the empirical distributions generated by no-regret dynamics in a Bayesian game accumulate on the set of BCCE, not on the smaller set of BCE. A seller who faces algorithmic bidders and wants a lower bound on revenue that holds for every outcome to which those algorithms can converge must therefore minimize over BCCE. This paper carries out an exercise similar to that of [Bergemann, Brooks](#)

¹See [Lomys and Magnolfi \(2025\)](#) for an econometric framework built on this weaker restriction, and [Nekipelov, Syrgkanis and Tardos \(2015\)](#) for inference from no-regret bidders.

and Morris (2017) for the first-price auction, with BCCE in place of BCE.

I define the **coarse revenue guarantee** (CRG) as the minimum expected revenue across all information structures and all BCCE, and I ask four questions. How large is it in the first-price auction, and which equilibrium attains it? How much does the seller lose by facing coarsely rational bidders rather than interim rational ones?² What can the seller do to raise it? And how do the results change across settings, namely symmetric and asymmetric priors, other auction formats, and an unknown value distribution?

Characterization through identical play The first contribution is a tractable characterization built on a structural result. In the first-price auction, from any BCCE one can construct another BCCE with weakly lower revenue in which every buyer submits the same bid with probability one. The worst case is then an *identical play*: bidders do not use their private information and all submit the same bid. The construction works because two effects act in the same direction. First, revenue in a first-price auction is the highest bid, so replacing a dispersed bid profile with a common bid that is no higher than the highest bid in the profile lowers revenue. This effect holds under any notion of rationality. Second, the replacement can be arranged so that the distribution of the *highest competing bid* faced by each buyer is unchanged, and this is what preserves the equilibrium constraints. A coarse deviation is a single fixed bid, and the profitability of a fixed bid depends on one’s rivals only through the distribution of the best bid among them. Holding that distribution fixed while lowering the price leaves each buyer’s payoff from deviating unchanged and raises their payoff from obeying.

The structure of this worst case is where the coarse analysis departs from the BCE benchmark. In Bergemann, Brooks and Morris (2017), the worst case is supported by an information structure in which bidders learn something about their rank and play symmetrically *conditional* on that information. Under coarse rationality, the worst case is instead unconditional identical play. Bidders who cannot condition a deviation on their signal can be induced to submit a common bid that does not depend on the

²The word *interim* refers to timing and is used throughout as the counterpart of *coarse*. In a Bayes correlated equilibrium a bidder decides whether to deviate after seeing their information, so the constraint is checked at the interim stage, conditional on the signal and the recommended bid. A coarse deviation, by contrast, is one fixed bid chosen before any information arrives, and its constraint averages over all states, which makes it an ex ante requirement. Section 3 gives the formal definitions.

signal.

The reduction also turns an N -dimensional problem into a one-dimensional one. The seller’s worst case no longer requires a search over joint distributions of values, signals, and N -dimensional bid profiles, since it is determined by the joint distribution of the value v and a single common bid c . Theorems 1 and 6 solve this problem in closed form for an arbitrary value distribution. In the worst case, buyers bid zero when the value is below a threshold, so the good is sold at a price of zero, and above the threshold they submit a common bid that increases with the value. The surplus from the zero-bid region is what keeps buyers from deviating to a slightly higher bid. The guarantee is strictly positive for every non-degenerate prior and strictly increasing in the number of buyers.

The cost of coarseness Because every BCE is a BCCE, the coarse guarantee is weakly below the interim guarantee of Bergemann, Brooks and Morris (2017). Proposition 4 shows that the inequality is strict at every finite N , so no BCE attains the coarse minimum. One way to see why is that the coarse worst case has a structure that interim obedience rules out: a bidder who is told to bid zero knows that the value is low and would gain by bidding slightly more. The gap is large when the number of bidders is small.

Nevertheless, Theorem 2 shows that this gap closes as N grows. The coarse constraint is satisfied by lowering revenue until each buyer’s equilibrium payoff is at least the payoff from every fixed deviation, and with N buyers sharing the surplus, that payoff is only $1/N$ of the total. As N grows, the revenue reduction needed to satisfy the coarse constraint goes to zero, and the two guarantees coincide in the limit. Proposition 3 shows, however, that the coarse guarantee converges more slowly. It reaches the common limit only at rate $\log N/N$, compared with $1/N$ for the interim guarantee, so the cost of coarseness $RG_N - CRG_N$ also vanishes at rate $\log N/N$.

What the seller could do The next question is what the seller can do to increase the coarse guarantee. One natural instrument is a reserve price, under which the good is withheld if the highest bid is below the reserve. Section 6 shows that the identical play reduction still holds with a reserve price once the pooling construction is modified, so the same tools apply and the guarantee can be characterized for every reserve below a bound given in Theorem 3.

Symmetric and asymmetric priors The main analysis assumes pure common values. The interim characterization of [Bergemann, Brooks and Morris \(2017\)](#) covers symmetric priors with arbitrary correlation, so it is natural to ask what changes when values may differ across buyers, and [Section 7](#) takes up this question. Identical play then need not be without loss. When values differ, the adversary who aims to reduce the seller’s revenue has an additional instrument. Through an arbitrarily small bid premium, it can keep the price common and still allocate the good to the buyer with the highest value, and the obedience constraints continue to hold. Nevertheless, [Theorem 4](#) shows that the guarantee for any exchangeable prior is bounded below by the fixed point of [Theorem 1](#) with the mean value replaced by the expected highest value. The bound depends on the prior only through each buyer’s own marginal and the expected highest value, so it extends to asymmetric priors with a separate set of constraints for each buyer, and [Theorem 5](#) gives the resulting lower bound.

Ranking and equivalence across auctions [Section 8](#) compares the first-price auction with the other standard auctions, following [Bergemann, Brooks and Morris \(2019\)](#). The comparison rests on one observation. Under identical play, every standard auction allocates by the same tie-breaking rule and charges the same price, so the outcome that attains the first-price guarantee is available in every format, and formats differ only in what a single deviator pays after outbidding the others. If the deviator pays their own bid, the identical play equilibria of the first-price auction are equilibria of the format. Its guarantee restricted to identical play is then the first-price guarantee, and its full guarantee is at most that. When the deviator pays the others’ bid instead, as in the second-price and English auctions, outbidding them does not raise the price paid, pooling at a low price cannot be sustained, and an asymmetric equilibrium yields a guarantee of zero. The first-price auction has the highest guarantee in both cases, so a seller designing for no-regret bidders is led to the same format as one designing for Bayesian bidders.

Unknown value distribution Finally, [Section 9](#) considers a seller who knows only the mean of the value distribution and an upper bound on the value, and who wants a lower bound on revenue over BCCE. I first show that the coarse revenue guarantee is monotone in the convex order: a mean-preserving spread of a prior is weakly worse for the seller. Because the binary distribution on $\{0, \bar{v}\}$ is the most

spread-out distribution on $[0, \bar{v}]$ with a given mean, it is the worst-case prior, and its guarantee is a lower bound that holds for every prior with that mean. I then solve for this bound in closed form.

The remainder of the paper is organized as follows. Section 2 discusses the related literature, and Section 3 introduces the model and the two equilibrium notions. Section 4 establishes the identical play reduction, which Section 5 uses to characterize the coarse revenue guarantee, its limit, and its rate of convergence. Section 6 studies reserve prices. Section 7 extends the lower bound to symmetric and asymmetric priors, Section 8 compares the guarantee across standard auctions, and Section 9 treats an unknown value distribution. Section 10 concludes. Omitted proofs are in the Appendix and the Supplementary Appendix.

2 Related Literature

This paper contributes to three strands of literature: first-price auctions, robust revenue guarantees, and the connection between Bayes coarse correlated equilibrium and no-regret learning.

First, this work builds on the literature on the first-price auction. Earlier studies, such as Vickrey (1961), Milgrom and Weber (1982), Battigalli and Siniscalchi (2003), Dekel and Wolinsky (2003), Fang and Morris (2006), and Azacis and Vida (2015), relied on specific assumptions about the information structure. More recently, Bergemann, Brooks and Morris (2017) allowed for arbitrary information structures but restricted attention to Bayes correlated equilibrium. This paper characterizes the coarse revenue guarantee under the weaker notion of BCCE and develops coarse counterparts of the ranking and equivalence results of Bergemann, Brooks and Morris (2019). The closest earlier work is Feldman, Lucier and Nisan (2016), who study correlated and coarse equilibria of single-item auctions. They work in the full-information model. I minimize revenue over all information structures and characterize the minimizing equilibrium exactly.

Second, this paper contributes to the literature on revenue guarantees, including Bergemann, Brooks and Morris (2017), Bergemann, Brooks and Morris (2019), Du (2018), and Brooks and Du (2021), by studying these guarantees beyond BCE. Work on the price of anarchy (Syrgkanis and Tardos, 2013; Jin and Lu, 2023) gives related guarantees for welfare. This paper is concerned with revenue.

Lastly, this work contributes to the literature on BCCE and learning in games. Coarse correlated equilibrium was introduced by [Moulin and Vial \(1978\)](#). It has received relatively little attention in economics, perhaps because its set of deviations is small. The use of no-regret algorithms as bidders has renewed interest in it. [Hartline, Syrgkanis and Tardos \(2015\)](#) show that no-regret dynamics in Bayesian games converge to the set of BCCE, building on complete-information results that link regret minimization to correlated and coarse correlated equilibria ([Hart and Mas-Colell, 2000](#); [Blum et al., 2008](#)). A growing literature studies auctions in which some or all participants are learning algorithms: [Nekipelov, Syrgkanis and Tardos \(2015\)](#) on inference from no-regret bidders, [Braverman et al. \(2018\)](#) and [Deng, Schneider and Sivan \(2019\)](#) on a seller’s ability to exploit them, [Kolumbus and Nisan \(2022\)](#) on auctions between learning agents, and [Lomys and Magnolfi \(2025\)](#) on structural estimation under no-regret restrictions. Relative to that literature, which typically fixes a learning protocol or an information structure, I take the set-valued approach of robust mechanism design and bound revenue over the entire BCCE set.

3 Model

3.1 Setup

Consider a single-unit auction with a set of risk-neutral buyers $\mathcal{N} = \{1, \dots, N\}$, where $N \geq 2$. The seller owns one indivisible good and values it at zero. The buyers have a pure common value for the good, denoted $v \in V = [0, \bar{v}] \subseteq \mathbb{R}_+$, distributed according to a probability measure $\mu \in \Delta(V)$ with mean $m := \mathbb{E}_\mu[v]$ and cdf F . [Section 7](#) treats symmetric and asymmetric priors, under which values may differ across buyers.

Assumption 1. *μ admits a density f that is continuous and positive on $[0, \bar{v}]$, and each buyer’s action set is $A_i = [0, \bar{v}]$, so that $A = \prod_i A_i = [0, \bar{v}]^N$.*

I impose the density assumption to simplify the exposition. Among the results, only [Proposition 3](#) relies on it, through the continuity and positivity of f at 0. Under [Assumption 1](#), F is continuous and strictly increasing on $[0, \bar{v}]$, and F^{-1} denotes its inverse. [Section 9](#) extends the characterization to arbitrary priors, including atomic ones, which is needed there because the worst-case prior among distributions with a given mean is binary.

In the first-price auction, a bid above $\bar{v}$ makes a loss whenever it wins, so restricting bids to $[0, \bar{v}]$ is without loss. I state the restriction explicitly because it may matter in Section 8, where the set of allowed deviations is what distinguishes one format from another.

In the first-price auction (FPA), the highest bidder wins. If there are ties, the good is allocated with equal probability among the tied bidders. Formally, the allocation rule $q_i(a)$ for buyer i is

$$q_i(a) = \begin{cases} 1, & \text{if } a_i > \max_{j \neq i} a_j, \\ \frac{1}{|\{k \in \mathcal{N} : a_k = \max_j a_j\}|}, & \text{if } a_i = \max_j a_j, \\ 0, & \text{otherwise,} \end{cases}$$

and the winner pays their own bid. Buyer i 's ex-post utility given the common value v and the action profile a is

$$u_i(v, a) = q_i(a)(v - a_i).$$

Since the good is always allocated, the buyers' payoffs add up to the value minus the price:

$$\sum_i u_i(v, a) = v - \max_i a_i \quad \text{for every } (v, a). \quad (1)$$

Anything that lowers revenue therefore raises the buyers' total surplus by the same amount.

3.2 Information and Two Notions of Obedience

The buyers' correlated information about the common value is described by an *information structure* $I = (S, \theta)$, where S_i is a measurable space of signals for buyer i , $S = \prod_i S_i$, and $\theta \in \Delta(V \times S)$ is the joint distribution of values and signals. Let $\mathcal{I}(\mu)$ be the set of information structures whose value marginal is the prior, $\text{marg}_V \theta = \mu$. The seller does not know I , and the guarantee is computed over all $I \in \mathcal{I}(\mu)$.

Given I , a *decision rule* is a measurable map $b : S \rightarrow \Delta(A)$. Equivalently, a mediator observes the signal profile s and privately recommends a bid a_i to each

buyer i . A decision rule induces the joint distribution

$$\sigma_b(dv, ds, da) = b(da | s) \theta(dv, ds) \in \Delta(V \times S \times A),$$

and buyer i 's expected utility is $U_i(I, b) := \int u_i(v, a) \sigma_b(dv, ds, da)$.

The two solution concepts I compare differ in one respect: *how much a buyer is allowed to know when choosing a deviation*. I state both in turn.

Definition 1 (Bayes Correlated Equilibrium). *A decision rule b is a Bayes correlated equilibrium of I if for every buyer i and every measurable map $\xi_i : S_i \times A_i \rightarrow A_i$,*

$$\int u_i(v, a) \sigma_b(dv, ds, da) \geq \int u_i(v, \xi_i(s_i, a_i), a_{-i}) \sigma_b(dv, ds, da).$$

Write $\mathcal{BCE}(I)$ for the set of BCE of I .

Definition 1 is the standard obedience formulation of [Bergemann and Morris \(2016\)](#). The map ξ_i may choose a different deviation for each signal s_i and each recommended bid a_i , so the recommendation must be optimal conditional on both.

Definition 2 (Bayes Coarse Correlated Equilibrium). *A decision rule b is a Bayes coarse correlated equilibrium of I if for every buyer i and every constant deviation $a'_i \in A_i$,*

$$\int u_i(v, a) \sigma_b(dv, ds, da) \geq \int u_i(v, a'_i, a_{-i}) \sigma_b(dv, ds, da).$$

Write $\mathcal{BCCCE}(I)$ for the set of BCCE of I .

In a BCCE, by contrast, the buyer considers only fixed-action deviations. No-regret learning dynamics converge to this set rather than to the smaller set of BCE ([Hartline, Syrgkanis and Tardos, 2015](#); [Blum et al., 2008](#)). In the language used throughout the paper, Definition 1 imposes *interim* obedience, which is evaluated after the buyer sees their information and therefore conditions on the signal. The coarse obedience of Definition 2 is an *ex ante* requirement, averaged over signals.

Because a constant map is a particular deviation map ξ_i , we have $\mathcal{BCE}(I) \subseteq \mathcal{BCCCE}(I)$ for every I , so BCCE is the more permissive concept. The following example illustrates the difference.

Example 1 (A BCCE that is not a BCE). Let $N = 2$ and let v take the values 0 and 1 with equal probability. Suppose both buyers observe v and the mediator recommends

the common bid $c = v/3$, so that both bid 0 when $v = 0$ and both bid $1/3$ when $v = 1$. Each buyer's expected payoff is $\frac{1}{2} \cdot \frac{1}{2} \cdot (1 - \frac{1}{3}) = 1/6$. A buyer deviating to the fixed bid $\delta \in (0, 1/3)$ wins only when $v = 0$, earning $\frac{1}{2}(0 - \delta) < 0$. A buyer moving to $\delta \in (1/3, 1)$ always wins and earns $\frac{1}{2}(0 - \delta) + \frac{1}{2}(1 - \delta) = \frac{1}{2} - \delta < \frac{1}{6}$. A buyer deviating to exactly $1/3$ wins when $v = 0$ and ties when $v = 1$, earning $\frac{1}{2}(0 - \frac{1}{3}) + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{2}{3} = 0$. Hence no fixed deviation earns more than $1/6$, and the profile is a BCCE. It is not a BCE, however. Conditional on the recommendation $c = 1/3$, the buyer knows that $v = 1$, and for small $\epsilon > 0$ a deviation to $1/3 + \epsilon$ wins the good and earns more than following the recommendation.

Example 1 illustrates the mechanism behind the paper's results. Under coarse rationality, revenue can be lowered by equilibria in which bidders would gain from deviating in the states where the good is valuable but are deterred by the losses they would incur in the states where it is not. The low bid can be sustained because the coarse constraint averages the deviation payoff across states, an averaging that interim obedience does not allow.

Remark 1 (Between coarse and interim obedience). The two definitions are the extreme cases of a family, ordered by how finely a deviation may condition on information. The coarse deviation of Definition 2 ignores the signals, and Definition 1 lets a deviation depend on the buyer's signal and recommendation. Contextual algorithms take value-relevant features as input (so they use only part of the information) and yield a solution concept between BCE and BCCE. As a result, the per-context guarantee lies between the two, and the coarse revenue guarantee is a valid lower bound for it.

With the two equilibrium notions in place, let $R(I, b) := \int \max_i a_i \sigma_b(dv, ds, da)$ be expected revenue. The *revenue guarantee* and the *coarse revenue guarantee* under the prior μ are

$$RG(\mu) := \inf_{I \in \mathcal{I}(\mu)} \inf_{b \in BCE(I)} R(I, b), \quad CRG(\mu) := \inf_{I \in \mathcal{I}(\mu)} \inf_{b \in BCCE(I)} R(I, b),$$

and since minimizing over a larger set yields a weakly smaller value, $CRG(\mu) \leq RG(\mu)$ for every prior.

3.3 A Reduced Notation for the Coarse Problem

Definition 2 has a property that Definition 1 lacks. Since a coarse deviation does not condition on the signal, the signals enter the constraint only through the joint distribution they induce over values and actions. The information structure can therefore be replaced, without loss, by a joint distribution $\sigma \in \Delta(V \times A)$ whose value marginal is μ .

Lemma 1 (Reduced formulation). *A distribution $\sigma \in \Delta(V \times A)$ with $\text{marg}_V \sigma = \mu$ is the value-action marginal of some σ_b with $I \in \mathcal{I}(\mu)$ and $b \in \mathcal{BCC}\mathcal{E}(I)$ if and only if, for every buyer i and every $a'_i \in A_i$,*

$$\int_{V \times A} u_i(v, a) d\sigma(v, a) \geq \int_{V \times A} u_i(v, a'_i, a_{-i}) d\sigma(v, a). \quad (2)$$

Proof. Necessity holds because both sides of Definition 2 depend on σ_b only through its $V \times A$ marginal. For sufficiency, let each buyer's signal be the bid that σ recommends to them: take $S_i = A_i$, let θ be σ itself, viewed as a law on $V \times S$, and let $b(s)$ be the point mass at s . Then $I \in \mathcal{I}(\mu)$, the induced law of (v, a) is σ , and the BCCE constraint of Definition 2 is (2). $\square$

Write $\mathcal{BCC}\mathcal{E}(\mu)$ for the set of σ satisfying (2), and $R(\sigma) = \int \max_i a_i d\sigma$. Then

$$CRG(\mu) = \inf_{\sigma \in \mathcal{BCC}\mathcal{E}(\mu)} R(\sigma),$$

and any σ attaining this infimum is called a *min-R BCCE*.

I do not assume that such a σ exists. Because the payoff functions u_i are discontinuous at ties, the map $\sigma \mapsto \int u_i d\sigma$ is not continuous in the weak-* topology, and $\mathcal{BCC}\mathcal{E}(\mu)$ need not be weak-* closed, so the existence of a minimizer does not follow from compactness. Every argument below is therefore constructive. Each step maps a given BCCE to another BCCE with weakly lower revenue, and I establish the existence of a minimizer at the end by construction.

Ties also require a separate argument, since the payoff functions are discontinuous there. Lemma 2 in Appendix A shows that for an identical play distribution it is enough to check the coarse constraint at bids that the buyers do not submit with positive probability, and these bids are dense. The same lemma gives the form of the constraint that every lower bound below uses.

4 The Identical Play Reduction

Minimizing revenue directly over $\mathcal{BCC}\mathcal{E}(\mu)$, a set of distributions over N -dimensional bid profiles, may not be tractable. This section shows that it is without loss of generality to restrict the minimization to a one-dimensional family. Call σ an *identical play* distribution if it puts probability one on profiles in which all buyers submit the same bid, $a_1 = \dots = a_N$. Call a distribution over (v, a) *exchangeable* if it is invariant under every permutation of the buyers' bids: for every bijection ρ of $\{1, \dots, N\}$, writing $(a_\rho)_i = a_{\rho(i)}$, the distribution is invariant under $(v, a) \mapsto (v, a_\rho)$.

Proposition 1 (Identical Play Reduction). *For every $\sigma \in \mathcal{BCC}\mathcal{E}(\mu)$ there is an exchangeable identical play $\gamma \in \mathcal{BCC}\mathcal{E}(\mu)$ with $R(\gamma) \leq R(\sigma)$, and the inequality is strict unless σ already places probability one on profiles whose two highest bids coincide. Consequently*

$$CRG(\mu) = \inf \{R(\gamma) : \gamma \in \mathcal{BCC}\mathcal{E}(\mu), \gamma \text{ identical play}\}.$$

The formal argument is in Appendix A. The main idea is that in the first-price auction, price competition comes from the buyers with the highest and the second-highest bids. To check the obedience constraint, a buyer needs only the distribution of the highest bid among the others. In any bid profile, the winner faces the second-highest bid and each of the other $N - 1$ buyers faces the highest bid, while the remaining bids play no role. A common bid equal to the highest bid with probability $(N - 1)/N$ and to the second-highest bid with probability $1/N$ therefore gives every buyer the competition that a randomly chosen buyer faced before, so each buyer's payoff from a fixed deviation equals the buyers' average payoff from that deviation in the original equilibrium. Revenue falls, so each buyer's payoff from obeying is at least the original average, and the obedience constraints still hold. This revenue reduction is unavailable only if the highest and the second-highest bids coincide with probability one. Since every buyer submits the common bid, the new BCCE is an identical play.

With this reduction in hand, the remaining problem is to choose the joint distribution of the value v and a single common bid c .

5 Characterization

5.1 The Representative-Bidder Program

Fix an identical play distribution $\gamma \in \mathcal{BCC}\mathcal{E}(\mu)$, and write c for the common bid and $R = \mathbb{E}[c]$ for expected revenue. Since all N bids coincide, every buyer wins with probability $1/N$ and pays c when they do, so each buyer's equilibrium payoff is

$$K := \frac{m - R}{N}, \quad (3)$$

the per-buyer share of the surplus not collected by the seller. I refer to K as the buyers' *rent*. A buyer who instead commits to the fixed bid δ wins whenever $c < \delta$ and shares when $c = \delta$, so coarse obedience requires, for every $\delta \in [0, \bar{v}]$,

$$K \geq \mathbb{E}\left[(v - \delta)\left(\mathbb{1}\{c < \delta\} + \frac{1}{N}\mathbb{1}\{c = \delta\}\right)\right]. \quad (4)$$

Setting $\delta = 0$ shows $K \geq 0$: revenue can never exceed the expected value.

Before stating the main result, I note two properties of the equilibrium constraint.

First, lowering revenue relaxes the constraints. By (3), lowering R raises K , which appears on the left-hand side of every constraint (4). The guarantee is then related to a K that makes the equilibrium constraint binding.

Second, a small deviation wins only in states where everyone else bids zero. Since each buyer earns K by following the recommendation, such a deviation is unprofitable as long as it earns less than K . Buyers can therefore all bid zero at low values, where they receive the good at a price of zero, and positive bids begin only above a value threshold. The following definition names the value at which the bidding region begins.

Definition 3 (Threshold and worst-case bidding rule). *Let $\hat{v}$ denote the pooling threshold: in the worst-case equilibrium, buyers bid zero whenever $v < \hat{v}$ and bid a strictly increasing amount above it. Write $H(v)$ for the common bid submitted at value $v \geq \hat{v}$.*

5.2 Main Result

Theorem 1. *Let Assumption 1 hold. The coarse revenue guarantee CRG and the threshold $\hat{v}$ are the unique solution of the system*

$$\frac{m - CRG}{N} = \int_0^{\hat{v}} v f(v) dv, \quad (5)$$

$$CRG = \int_0^{\bar{v}} H(v) f(v) dv, \quad (6)$$

where $H(v) = 0$ for $v < \hat{v}$ and, for $v \geq \hat{v}$,

$$H(v) = \frac{1}{F(v)} \int_{\hat{v}}^v z f(z) dz = \mathbb{E}[z \mid \hat{v} \leq z \leq v] \cdot \frac{F(v) - F(\hat{v})}{F(v)}. \quad (7)$$

The guarantee is strictly positive and can be attained: the identical play distribution with common bid $c = H(v)$ is a BCCE achieving it.

Appendix B contains the proof, which has three main steps. A *lower bound* shows that no identical play BCCE yields revenue below the value given by the system. It comes from combining the obedience constraint with a rearrangement inequality, rather than from assuming that constraints bind. I refer to the resulting sequence of inequalities (Lemma 5) as the *chain*. *Sufficiency* verifies that the candidate rule (7) deters every fixed deviation, including deviations above the top of the bid range. The last step is the *uniqueness* of the solution to (5) and (6). This step is not immediate, since both sides of the system move in the same direction as $\hat{v}$ rises, and it follows from a concavity argument given in the appendix.

The fixed-point form For later use, I record an equivalent form of the system, written in the quantile variable $q = F(v)$. Let

$$\Phi_\mu(q) := \int_0^q F^{-1}(t) dt = \int_0^{F^{-1}(q)} z f(z) dz, \quad q \in [0, 1], \quad (8)$$

the expected value collected on the lowest q quantiles of the prior. It is convex and strictly increasing, with $\Phi_\mu(0) = 0$ and $\Phi_\mu(1) = m$. Define the revenue functional

$$\mathcal{R}_\mu(K) := \int_0^1 \frac{(\Phi_\mu(q) - K)^+}{q} dq, \quad K \in [0, m]. \quad (9)$$

The system of (5) and (6) is equivalent to the single equation

$$m - NK = \mathcal{R}_\mu(K) \quad (10)$$

in the rent: the unique solution is $K^* = \int_0^{\widehat{v}} v f(v) dv$, the guarantee is $CRG = m - NK^* = \mathcal{R}_\mu(K^*)$, and the worst-case bid is $c = \psi(F(v)) = H(v)$, where

$$\psi(q) := \frac{(\Phi_\mu(q) - K^*)^+}{q}. \quad (11)$$

Appendix B proves the theorem in this form. The functional $\mathcal{R}_\mu$ is also used in the reserve-price analysis of Section 6, in the bounds of Section 7, and in the convex-order comparison of Section 9.

Equation (5) is the obedience constraint for a bid just above zero. Its right-hand side is the expected value collected on the pooling region, which is also the payoff from deviating to such a bid. The rent K must be large enough to make that deviation unattractive, and in the worst case it is exactly large enough.

Equation (7) describes the bidding rule above the threshold. The common bid at value v is the expected value conditional on lying in the bidding region and below v , shaded by the probability that a win comes from the bidding region rather than from the pooling region. Comparing it with the limiting BCE bidding rule $G(v) = \mathbb{E}[z \mid z \leq v]$ shows the source of the gap:

$$G(v) - H(v) = \frac{K}{F(v)} \quad \text{for } v \geq \widehat{v}.$$

Above the threshold, the coarse bidding rule is the limiting interim rule minus $K/F(v)$. The gap between CRG and the limiting guarantee of Theorem 2 is therefore the integral of this term against f over the bidding region, plus the integral of G against f over the pooling region.

Example 2 (Uniform values). Let $\mu = U[0, 1]$, so $m = 1/2$ and $F(v) = v$. Then $H(v) = (v^2 - \widehat{v}^2)/(2v)$, and the system reduces to the single equation

$$\frac{1}{2} - \frac{N\widehat{v}^2}{2} = \frac{1}{4} - \frac{\widehat{v}^2}{4} + \frac{\widehat{v}^2 \log \widehat{v}}{2}.$$

With two buyers the threshold is $\widehat{v} \approx 0.6730$ and $CRG \approx 0.0471$, so the seller collects

less than 10% of the good's expected value. Since $F(\hat{v}) = \hat{v}$ for the uniform distribution, the pooling region has probability 0.673: in the worst case, both buyers bid zero with probability 0.673, and the good is then sold at a price of zero. Conditional on a positive price being paid, the average price is 0.144. With ten buyers, $CRG \approx 0.1916$ and the pooling probability falls to 0.248.

Proposition 2 (Comparative statics in N). *$CRG(\mu)$ is strictly increasing in N , and the rent K is strictly decreasing in N .*

The guarantee increases in N for a reason that seems specific to coarse rationality. The rent that deters a fixed deviation is a per-buyer quantity: each buyer receives K , and by (3) the total is $NK = m - R$. With more buyers, each one's share of a given amount of uncollected surplus is smaller, so a given level of revenue leaves each buyer with a smaller equilibrium payoff, and the revenue reduction needed to satisfy the constraints is smaller.

5.3 Asymptotic Convergence and Its Rate

Theorem 2. *Under Assumption 1,*

$$\lim_{N \rightarrow \infty} CRG(\mu) = \int_0^{\bar{v}} G(v) f(v) dv = \lim_{N \rightarrow \infty} RG(\mu), \quad G(v) = \mathbb{E}[z \mid z \leq v].$$

The proof is in Appendix B. Intuitively, as N grows, the rent K vanishes at rate $1/N$, the pooling threshold $\hat{v}$ falls to zero, and by the identity $G - H = K/F$ the coarse bidding rule converges pointwise to the limiting interim rule G characterized by Bergemann, Brooks and Morris (2017).

A second, heuristic argument for the same convergence connects the result to the information structure of the benchmark. The worst case in Bergemann, Brooks and Morris is supported by an information structure in which each bidder learns only whether they hold the highest as-if-value signal, together with a summary of the remaining as-if-value signals. As N grows, that information structure converges to the value distribution itself, so conditioning on it adds no information, and the two notions of obedience impose the same restriction in the limit.

The next question is the rate of convergence. For pure common values, Bergemann, Brooks and Morris (2015), the working paper version of Bergemann, Brooks

and Morris (2017), characterize the minimum-revenue BCE. In the worst case the winning bid is a deterministic function of the value, and the bid and the revenue are

$$\beta_N(v) = \frac{\int_0^v z dF^{\frac{N-1}{N}}(z)}{F^{\frac{N-1}{N}}(v)}, \quad RG_N(\mu) = \int_0^{\bar{v}} \beta_N(v) f(v) dv, \quad (12)$$

where $\beta_N(v)$ is the conditional mean of the value under the tilted distribution $F^{(N-1)/N}$.³ For the uniform prior, $\beta_N(v) = \frac{N-1}{2N-1} v$ and $RG_N = \frac{N-1}{2(2N-1)}$, which equals $1/6$ at $N = 2$ and increases to $1/4$. In this example the interim guarantee converges to the common limit at rate $1/N$. The next result shows that the coarse guarantee converges at the slower rate $\log N/N$.

Proposition 3 (Rate of convergence). *Let Assumption 1 hold and let R_∞ denote the limit in Theorem 2. Then $0 \leq R_\infty - RG_N \leq 2\bar{v}/(N-1)$, while*

$$R_\infty - CRG_N = \frac{(m - R_\infty) \log N}{2N} (1 + o(1)).$$

Consequently the cost of coarseness satisfies

$$RG_N - CRG_N = \frac{(m - R_\infty) \log N}{2N} (1 + o(1)) :$$

the interim guarantee converges at rate $1/N$, the coarse guarantee at rate $\log N/N$, and the gap between them closes at $\log N/N$.

The proof is in Supplementary Appendix S.1. The gap $R_\infty - CRG_N$ equals the rent K times $\log(1/q_K)$, up to a term of order $1/N$, and K is of order $1/N$ for every prior. The rate therefore depends on how fast the pooling quantile q_K goes to zero, and this is the one place where the continuity and positivity of f at 0 in Assumption 1 are used. Near zero, $\Phi_\mu(q) \approx q^2/(2f(0))$, hence $q_K \approx \sqrt{2f(0)K}$, and the square root produces the factor $\frac{1}{2}$ in the rate. If instead f behaves like $x^{\alpha-1}$ near zero for some $\alpha > 0$, so that $f(0)$ is zero or infinite, then $q_K \approx K^{\alpha/(\alpha+1)}$, the rate is still $\log N/N$, and the factor $\frac{1}{2}$ is replaced by $\alpha/(\alpha+1)$.

Table 1 and Figure 1 illustrate these rates for the uniform prior at matched market

³See Bergemann, Brooks and Morris (2015) for the common value case. In the as-if reading, each of N independent values has marginal $F^{1/N}$, so that the highest of them is distributed as F and the highest rival value as $F^{(N-1)/N}$. Then β_N is the symmetric equilibrium bid of that model, the expected highest rival value conditional on winning.

N	2	3	5	10	20	50	100
Interim guarantee RG_N	0.1667	0.2000	0.2222	0.2368	0.2436	0.2475	0.2487
Coarse guarantee CRG_N	0.0471	0.0939	0.1453	0.1916	0.2178	0.2353	0.2419
Cost of coarseness $RG_N - CRG_N$	0.1196	0.1061	0.0770	0.0452	0.0258	0.0122	0.0068
CRG_N below RG_N (%)	71.7	53.1	34.6	19.1	10.6	4.9	2.7
Pooling probability $F(\hat{v})$	0.673	0.520	0.377	0.248	0.168	0.103	0.072

Table 1: Interim and coarse revenue guarantees for $\mu = U[0, 1]$, both converging to $1/4$. The interim row is $RG_N = \frac{N-1}{2(2N-1)}$, computed from the [Bergemann, Brooks and Morris \(2017\)](#) characterization via (12). The comparison is at the same market size throughout.

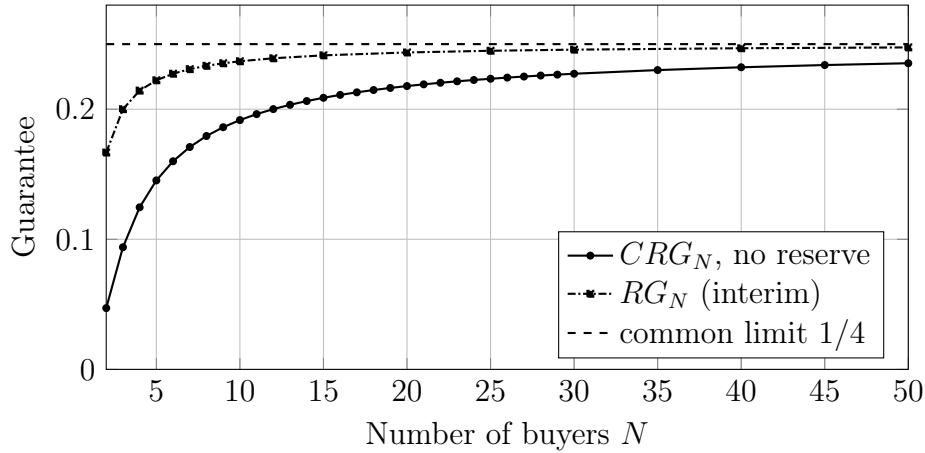


Figure 1: Interim and coarse revenue guarantees for $\mu = U[0, 1]$

sizes. With two bidders the coarse guarantee is under thirty percent of the interim guarantee, and with ten it is about eighty percent.

The strict inequality between RG_N and CRG_N is not specific to the uniform prior. The next result shows that it holds for every prior and every finite N , provided the interim minimum is attained as in [Bergemann, Brooks and Morris \(2017\)](#).

Proposition 4 (Strict coarseness gap). *Let Assumption 1 hold and suppose the infimum defining $RG_N(\mu)$ is attained by a BCE in which the two highest bids differ with positive probability, as in the construction of [Bergemann, Brooks and Morris \(2017\)](#). Then*

$$CRG_N(\mu) < RG_N(\mu) \quad \text{for every finite } N.$$

The proof, in Appendix B, uses only Proposition 1. In the minimum-revenue BCE of [Bergemann, Brooks and Morris](#), the winning bid is a deterministic function

of the value and the losing bids lie strictly below it, so the two highest bids differ. By Proposition 1, pooling such a distribution yields a BCCE with strictly lower revenue. The interim minimum is therefore not a coarse minimum, and the two guarantees differ.

6 Reserve Prices

When N is small, the worst case of Sections 4 and 5 sells the good at a price of zero with high probability. This section shows that a reserve price can raise the guarantee substantially.

Formally, fix a reserve price $r \in [0, \bar{v})$. The first-price auction with reserve r allocates the good to the highest bidder at the bid price if the highest bid is at least r , breaking ties uniformly, and withholds the good otherwise. Write $CRG_N(r)$ for the resulting coarse revenue guarantee.

Bids below r never win, so without loss of generality every bid is either 0, interpreted as staying out, or at least r . Since the good is no longer always allocated, equation (1) becomes

$$\sum_i u_i(v, a) = (v - a_{(1)}) \mathbb{1}\{a_{(1)} \geq r\}. \quad (13)$$

The pooling construction of Proposition 1 cannot be applied directly. Suppose exactly one buyer clears the reserve, so $a_{(1)} \geq r > a_{(2)}$. Pooling onto $a_{(2)}$ with probability $1/N$ would then withhold the good, and by (13) this removes the buyers' surplus together with the seller's revenue, which can tighten the obedience constraints. A modified construction still works: pool onto $\max\{a_{(2)}, r\}$ instead, so that the good is still sold, and switch to staying out only when the sale yields negative surplus, that is, when the winning bid exceeds the value.

Proposition 5 (Identical Play with a Reserve). *Fix $r \in [0, \bar{v})$. For every BCCE σ of the first-price auction with reserve r there is an exchangeable identical play BCCE γ with $R(\gamma) \leq R(\sigma)$, where the common bid takes values in $\{0\} \cup [r, \bar{v}]$.*

The proof is in Appendix C.

Under a reserve, buyers can no longer receive the good at a price of zero, but they have a less costly alternative: pooling at exactly r , where the good still trades and

the winner pays only the reserve. For values below r , however, buyers prefer not to buy at all. The worst case therefore involves up to two pools: an *opt-out pool* at zero, where the good is not sold, and a *reserve pool* at exactly r , where the good trades at the reserve. The opt-out pool occupies the lowest s quantiles, the reserve pool the next $q_0 - s$, and above q_0 the common bid is strictly increasing. The family is determined as follows. A bid just above the reserve wins against both pools outright, so obedience requires $K \geq \Phi_\mu(q_0) - r q_0$. The bidding rule must also start at or above the reserve, $(\Phi_\mu(q_0) - K)/q_0 \geq r$, which is the same inequality reversed. Hence, in the worst case,

$$K = \Phi_\mu(q_0) - r q_0. \quad (14)$$

The opt-out pool removes the value $\Phi_\mu(s)$ from the buyers' total surplus, so

$$NK = m - \Phi_\mu(s) - R, \quad R = r(q_0 - s) + \int_{q_0}^1 \frac{\Phi_\mu(q) - K}{q} dq, \quad (15)$$

Given the opt-out share s and the rent K , (14) places q_0 where the increasing part of the bidding rule crosses the reserve, so the bid at every quantile $q > s$ is $\max\{r, (\Phi_\mu(q) - K)/q\}$. Equations (14) and (15) therefore reduce to a single equation in the rent,

$$NK = m - \Phi_\mu(s) - \int_s^1 \max\left\{r, \frac{\Phi_\mu(q) - K}{q}\right\} dq. \quad (16)$$

By (15), revenue is $m - \Phi_\mu(s) - NK$, so for a given opt-out share the worst case uses the largest solution $K = \kappa_+(s)$ of (16) in $[0, m]$. Under the condition of Theorem 3 below, $\kappa_+(s) < m - r$, so the increasing part of the bidding rule does cross the reserve at some $q_0 < 1$. Call s *feasible* if $\kappa_+(s)$ exists and $\Phi_\mu(s) - rs \leq \kappa_+(s)$, which then says that the reserve pool $[s, q_0)$ is well defined and that the average value in it is at least r . Let

$$\widehat{V}_N(r) := \min \{ m - \Phi_\mu(s) - N\kappa_+(s) : s \text{ feasible} \}.$$

Theorem 3 (Reserve prices). *Let $r \in [0, m]$ satisfy $rF(r) - \Phi_\mu(F(r)) < (N - 1)(m - r)$, which is true for every $r \leq m(N - 1)/N$ and for every r in the example below. Then*

$$CRG_N(r) = \widehat{V}_N(r),$$

so the two-pool family is the worst case. Moreover, the guarantee is continuous at a

r	0.01	0.05	0.10	0.148	0.20
$CRG_2(r)$	0.0552	0.0812	0.1000	0.1056	0.1038
opt-out share s	0.051	0.253	0.464	0.621	0.664
pool boundary q_0	0.676	0.674	0.650	0.621	0.664

Table 2: The worst case across reserve levels, $\mu = U[0, 1]$, $N = 2$. The reserve pool has mass $q_0 - s$. It contains most of the pooled mass at small reserves, shrinks as the reserve grows, and vanishes at the reserve that maximizes the guarantee, $r \approx 0.148$, beyond which the worst case relies on opting out alone. Values are computed from (14) and (15).

zero reserve, $\lim_{r \downarrow 0} CRG_N(r) = CRG_N(0)$.⁴ For $r \geq m$ the guarantee is zero, since no participation is a BCCE.

The proof is in Appendix C, with the continuity claim in Supplementary Appendix S.2. The lower bound follows the chain of Theorem 1 with two changes. The bid quantile function is now bounded below by $\max\{r, (\Phi_\mu(q) - K)/q\}$ rather than by $(\Phi_\mu(q) - K)/q$, and equation (13) removes the value of the states in which buyers opt out, which by the rearrangement inequality is at least $\Phi_\mu(s)$. For a given opt-out share, the largest rent consistent with both bounds defines a member of the two-pool family, which attains the bound.

Example 3 (Uniform values with a reserve). Return to the setting of Example 2, where $\mu = U[0, 1]$, $N = 2$, and without a reserve $CRG_2 \approx 0.047$, sustained by both buyers bidding zero on the lowest two-thirds of values and receiving the good at a price of zero. Table 2 traces the worst case as the reserve grows. At $r = 0.01$ buyers bid r on nearly the entire former pooling region, almost no one opts out, and the guarantee rises continuously from $CRG_2(0)$, with slope about 0.84 at $r = 0$ (Supplementary Appendix S.2). As the reserve grows, pooling at the reserve becomes costly for the buyers and the worst case shifts toward opting out, so the share s rises. The guarantee is maximized at $r \approx 0.148$, the reserve at which the reserve pool vanishes. Below that level, part of the pooling region still bids r . A larger reserve lowers the probability of sale and does not tighten obedience further. At this reserve the guarantee is $CRG_2 \approx 0.106$, more than double the guarantee without a reserve.

⁴The characterization was also checked by solving a discretized version of the coarse program as a linear program, with neither identical play nor comonotonicity imposed. The linear program reproduces $\hat{V}_N(r)$ at every reserve tested.

The gain accrues gradually: reaching half of it requires $r \approx 0.041$.

7 Beyond Common Values: Symmetric and Asymmetric Priors

The analysis so far, with and without a reserve, assumes pure common values. The interim characterization of [Bergemann, Brooks and Morris \(2017\)](#) covers symmetric priors with arbitrary correlation, and this section asks how much of the coarse analysis extends to values that may differ across buyers. Let π be a prior on V^N that is exchangeable, that is, invariant under every permutation of the coordinates, with common per-buyer marginal μ . Buyer i 's payoff is $q_i(a)(v_i - a_i)$, and a BCCE is defined as in Lemma 1, now over joint laws of $(v_1, \dots, v_N, a_1, \dots, a_N)$ whose marginal on the value profile is π . Such a law is exchangeable if it is invariant under permuting values and bids together, $(v, a) \mapsto (v_\rho, a_\rho)$ with $(v_\rho)_i = v_{\rho(i)}$, for every bijection ρ of $\{1, \dots, N\}$. Pure common values are the coupling $v_1 = \dots = v_N$, and independent private values are the opposite extreme. Write $CRG(\pi)$ for the infimum of revenue over all BCCE under π . Under common values, Proposition 1 reduces this infimum to identical play, but the reduction does not hold in general when values differ. This section explains why and gives the lower bound that remains valid, first for symmetric priors and then for asymmetric ones.

The interim characterization holds for every symmetric prior, whatever the coupling of values across buyers. The identical play reduction over BCCE does not, and the reason is specific to coarse obedience. Under BCE, the correlation structure of the prior is irrelevant to the seller's problem, in the sense that the method of [Bergemann, Brooks and Morris \(2017\)](#) handles every coupling. Under BCCE it matters, and the contrast comes from how each solution concept constrains a bidder. In a BCE, obedience is checked conditional on the bidder's own recommendation, so nature can correlate each bidder's signal with that bidder's own value. This gives nature N separate instruments, one per bidder, and it is why the worst case of [Bergemann, Brooks and Morris](#) handles arbitrary couplings and is even allocatively efficient. A bidder's own bid, by contrast, appears nowhere in their coarse deviation payoff $\mathbb{E}[(v_i - \delta)\mathbb{1}\{\max_{j \neq i} a_j < \delta\}]$, so under BCCE the only thing that constrains buyer i is the correlation between v_i and the *other* buyers' bids.

Turning to the reduction, whether the identical play program yields the guarantee away from common values depends on whether the reduction of Proposition 1 holds, and in general it does not. Averaging over relabelings of the buyers still makes every buyer symmetric without changing revenue (Appendix D), but the pooling step fails. The pooling kernel of Lemma 3 allocates the good to a uniformly random buyer instead of to the top bidder. Under common values this changes nothing, since every buyer values the good equally. When values differ, it changes two things. First, if the top bidder has an above-average value, reassigning the good at random lowers the buyers' total surplus, so the pooled rent can fall rather than rise. Second, a deviator who outbids everyone except the top bidder is, by exchangeability, someone other than the top bidder, so the deviation payoff can change. Both corrections have the same sign, so pooling still works when the symmetrized distribution allocates against value but can fail when it allocates efficiently. With heterogeneous values the adversary lowers revenue by allocating efficiently: the good goes to the buyer with the highest value, the buyers' total surplus rises, and the additional surplus relaxes the obedience constraints, which permits lower bids.

Moreover, in the first-price auction the adversary can do this at an arbitrarily small cost in revenue. Fix a common price and give the buyer with the highest value a bid premium of ε above it. The premium decides the allocation and adds only ε to revenue, while every coarse deviation still faces, up to ε , the common price as its highest competing bid. The lower bound that remains valid follows the chain of Theorem 1, with the mean value replaced by the expected highest value. Let $m_N := \mathbb{E}_\pi[\max_i v_i]$ denote the expected highest value, so that $m \leq m_N \leq Nm$, and let $\tilde{K}$ be the unique solution in $(0, m]$ of

$$m_N - NK = \mathcal{R}_\mu(K), \quad (17)$$

which exists by the argument of Lemma 7 with m_N in place of m .

Theorem 4 (Symmetric priors). *Let π be exchangeable with per-buyer marginal μ satisfying Assumption 1. Then*

$$CRG(\pi) \geq m_N - N\tilde{K} = \mathcal{R}_\mu(\tilde{K}).$$

The proof is in Appendix D. The bound follows the chain of Theorem 1 with two

substitutions: the rearrangement inequality is applied to the pair (v_i, M_i) of a buyer's own value and highest competing bid, which uses only the marginal μ , and the payoff equation bounds the buyers' total surplus by the efficient surplus $m_N - R$ rather than by $m - R$. Theorem 4 gives a lower bound only. When $m_N > m$, identical play does not attain it, since a common bid allocates the good at random on the pooling event and the buyers' total surplus is then below the efficient surplus, and the premium construction above shows how the adversary can do better. Whether the bound is attained for a given prior is left open.

The proof uses the prior π only through buyer i 's own marginal, in the chain, and through m_N , in the surplus bound. Exchangeability is never used, so the theorem holds for every prior with common per-buyer marginal μ , whatever the coupling. When the marginals differ, the same two steps give a bound in which each buyer has their own chain. Let π be any prior on V^N with per-buyer marginals $\mu_1, \dots, \mu_N$, write $m_i := \mathbb{E}[v_i]$, define Φ_{μ_i} and $\mathcal{R}_{\mu_i}$ by (8) and (9) with μ_i in place of μ , and let

$$\mathcal{R}_{\mu_i}^{-1}(R) := \inf \{K \in [0, m_i] : \mathcal{R}_{\mu_i}(K) \leq R\}, \quad (18)$$

the smallest rent at which buyer i 's chain permits revenue R . Since $\mathcal{R}_{\mu_i}$ is continuous and strictly decreasing from $\mathcal{R}_{\mu_i}(0)$ to 0 on $[0, m_i]$, this is the inverse of $\mathcal{R}_{\mu_i}$ for $R \leq \mathcal{R}_{\mu_i}(0)$ and equals zero above.

Theorem 5 (Asymmetric priors). *Let π be a prior on V^N whose per-buyer marginals $\mu_1, \dots, \mu_N$ each satisfy Assumption 1, and let $m_N := \mathbb{E}_\pi[\max_i v_i]$. Then $\text{CRG}(\pi) \geq \tilde{R}$, where $\tilde{R}$ is the smallest $R \in [0, m_N]$ satisfying*

$$\sum_{i=1}^N \mathcal{R}_{\mu_i}^{-1}(R) \leq m_N - R. \quad (19)$$

When $\mu_i = \mu$ for every i , $\tilde{R} = m_N - N\tilde{K}$, so Theorem 4 is the case of identical marginals.

The proof is in Appendix D. Each buyer's chain says that revenue R requires buyer i 's rent to be at least $\mathcal{R}_{\mu_i}^{-1}(R)$, and the surplus bound says that the rents sum to at most $m_N - R$. A revenue level is possible only if the rents that every chain requires fit within the surplus that the allocation can generate, and $\tilde{R}$ is the smallest such level. In the symmetric case every chain requires the same rent, (19) reads

$N\mathcal{R}_\mu^{-1}(R) \leq m_N - R$, and its smallest solution corresponds to the fixed point (17). With different marginals the adversary can distribute the total rent unevenly, giving more to the buyers whose chains are harder to satisfy, and (19) accounts for this. As in the symmetric case, attainment would require a BCCE that allocates efficiently on the pooling event and makes every buyer's chain bind at the same revenue, and I do not pursue it here.

8 Ranking and Equivalence Across Auctions

I now return to common values and compare the first-price auction with other formats, following Bergemann, Brooks and Morris (2019). As in their framework, a *standard auction* is one that allocates the good to the highest bidder, breaking ties uniformly, and admits an equilibrium in monotonic pure strategies in the symmetric independent private values model. Losers pay nothing, and the winner pays $p(a)$, which I take to satisfy $p(a) \leq a_{(1)}$, with equality whenever $a_{(1)} = a_{(2)}$: the winner never pays more than their bid, and a winner who is tied with another bidder pays the tied bid. Let $\mathcal{M}$ denote this class. Within it, the pricing rules that matter are defined by what happens on profiles in which the $N - 1$ losing bids coincide. A standard auction has *own-bid pricing* if the winner pays $a_{(1)}$ whenever the other bids coincide, as in the first-price and Dutch auctions, and *competing-bid pricing* if the winner pays $a_{(2)}$ whenever the other bids coincide, as in the second-price and English auctions.

The comparison combines two facts, as in Bergemann, Brooks and Morris. First, identical play is an outcome that every standard auction can produce at the same revenue. Second, formats differ only in what a single deviator pays after outbidding the others.

Proposition 6 (Identical play is format-free). *Fix any identical play distribution with common bid c . In every standard auction the good is allocated uniformly among the N buyers, the winner pays c , and expected revenue is $\mathbb{E}[c]$. In every standard auction with own-bid pricing, moreover, the identical play BCCE are exactly those of the first-price auction, so the coarse revenue guarantee restricted to identical play equals $CRG(\text{FPA})$ in each of them.*

Proof. Under own-bid pricing, a buyer who deviates from an identical play distribution to the fixed bid δ loses if $\delta < c$, ties at price c if $\delta = c$, and wins at price δ if

$\delta > c$, exactly as in the first-price auction, and equilibrium payoffs coincide by the first part. Hence the coarse constraints (2) are the same in both formats, and the sets of identical play BCCE coincide. $\square$

Proposition 7 (Coarse revenue guarantee ranking). *The first-price auction has the highest coarse revenue guarantee among standard auctions with own-bid or competing-bid pricing. For every $M \in \mathcal{M}$ with own-bid pricing, $CRG(M) \leq CRG(\text{FPA})$, and for every $M \in \mathcal{M}$ with competing-bid pricing, in particular the second-price and English auctions,*

$$CRG(M) = 0 < CRG(\text{FPA}).$$

Proof. Under own-bid pricing, the identical play BCCE attaining $CRG(\text{FPA})$ in Theorem 1 is a BCCE of M with the same revenue, by Proposition 6, so $CRG(M) \leq CRG(\text{FPA})$. Under competing-bid pricing, consider the distribution in which one designated buyer bids $\bar{v}$ and all others bid zero, independently of the value. Revenue is zero. The designated buyer earns m and cannot improve, since any positive bid still wins at the price $p(\delta, 0, \dots, 0) = 0$. Any other buyer must bid $\bar{v}$ to contest the good, which ties the top bid and, since a winner who is tied pays the tied bid, means winning with some probability $\lambda \in (0, 1]$ at price $\bar{v}$, for a payoff of $\lambda(m - \bar{v}) \leq 0$, and the conclusion does not depend on how ties are broken. This is a BCCE with zero revenue, so $CRG(M) = 0$, and $CRG(\text{FPA}) > 0$ by Theorem 1. $\square$

9 Unknown Value Distribution

So far the seller has been assumed to know the prior. Suppose instead that the seller knows only the expected value $E = \mathbb{E}[v]$ and the upper bound $\bar{v}$ on the value, but not the shape of the distribution. The worst-case prior is then binary, which violates Assumption 1, so I first extend the characterization to priors without a density, although with heavier notation.

Theorem 6 (Beyond the density). *Let μ be any non-degenerate distribution on $[0, \bar{v}]$, and read the definitions (8), (9), and (11) with the quantile function $F^{-1}(t) = \inf\{x : F(x) \geq t\}$ in place of the inverse. Then (10) has a unique solution $K^* \in (0, m)$, and $CRG(\mu) = m - NK^* = \mathcal{R}_\mu(K^*) > 0$. The infimum is attained by the identical play distribution $v = F^{-1}(U)$, $c = \psi(U)$, with U uniform on $[0, 1]$.*

In the attaining distribution, the uniform variable is the rank of the value. Away from the atoms of μ we have $U = F(v)$, so the coupling reduces to the deterministic bid $H(v) = \psi(F(v))$ of Theorem 1. Conditional on an atom of μ , however, U is uniform on the corresponding quantile interval, so the common bid is randomized continuously across it. The randomization is required by the constraints. Minimality requires the obedience constraint to bind at every deviation level, and a deviation to $\psi(q)$ with q inside an atom's interval must win exactly a fraction of the atom, which a single common bid cannot arrange. If the common bid were the same across the whole atom, a deviation just above it would win the entire atom at once and earn strictly more than the rent.

The proof of Theorem 1 in Appendix B is written for an arbitrary prior, so it proves Theorem 6 as well (Supplementary Appendix S.3). Neither the identical play reduction of Proposition 1 nor the characterization uses the density: the rearrangement inequality holds for arbitrary priors once its equality clause allows an atom to be split, the concavity lemma behind uniqueness does not use the density, and the attaining coupling randomizes the common bid at any atom of the prior, in the manner just described.

With the characterization extended to atomic priors, the relevant comparison across priors is the convex order. Write $\mu \preceq_{\text{cx}} \nu$ if $\int \phi d\mu \leq \int \phi d\nu$ for every convex function ϕ on $[0, \bar{v}]$, or equivalently if μ and ν have the same mean and ν is a mean-preserving spread of μ . I read $\mu \preceq_{\text{cx}} \nu$ as “ ν is riskier than μ .”

Proposition 8 (Convex order monotonicity). *If $\mu \preceq_{\text{cx}} \nu$ then $\text{CRG}(\nu) \leq \text{CRG}(\mu)$, for every N . Consequently, for every N ,*

$$\min_{\mu: \text{supp } \mu \subseteq [0, \bar{v}], \mathbb{E}_\mu[v] = E} \text{CRG}(\mu) = \text{CRG}(\underline{\mu}),$$

where $\underline{\mu}$ is the binary distribution with $\underline{\mu}(\{0\}) = 1 - E/\bar{v}$ and $\underline{\mu}(\{\bar{v}\}) = E/\bar{v}$.

The proof is in Supplementary Appendix S.3. A low price is sustained in the worst case because the bid increases with the value, so a buyer who deviates to a slightly higher bid wins mostly in states where the value is low, and the deviation is unprofitable despite the low price. A mean-preserving spread lowers the expected value on the lowest quantiles, $\Phi_\nu \leq \Phi_\mu$, which makes such deviations even less profitable. Lower bids then satisfy the obedience constraints, and revenue falls. Since the

binary distribution on $\{0, \bar{v}\}$ is the riskiest distribution on $[0, \bar{v}]$ with a given mean, it is the seller's worst case.

Corollary 1 (The binary guarantee). *Let $p_0 = 1 - E/\bar{v}$. For finite N , $CRG(\underline{\mu}) = E - N\bar{v}(q^* - p_0)$, where $q^* \in (p_0, 1)$ is the unique solution of $(N-1)(p_0 - q) = q \log q$. As $N \rightarrow \infty$,*

$$CRG(\underline{\mu}) \rightarrow E + (\bar{v} - E) \log \left(1 - \frac{E}{\bar{v}}\right). \quad (20)$$

Stating the result through the convex order, rather than through the extreme point alone, has three advantages. It holds at every N and not only in the limit. It also covers distributions with atoms, which matters because the minimizer is itself atomic. Finally, it gives a full comparative static rather than only the worst case, so a seller with partial information about dispersion can locate their guarantee between the extremes. For $E = 1/2$ and $\bar{v} = 1$, the ordering $U[0, 1] \preceq_{\text{cx}} \underline{\mu}$ corresponds to the reverse ordering of guarantees: 0.047 against 0.040 at $N = 2$, and 0.250 against 0.153 in the limit.

10 Conclusion

This paper characterizes the worst-case revenue of auctions when bidders are rational only in the coarse sense that no single fixed bid would have earned them more on average. No-regret learning leads to this restriction, and it seems well suited to algorithmic bidders. In the first-price auction the worst case is an identical play equilibrium, in which bidders do not use their private information and all submit the same bid, and this reduction yields a closed-form characterization of the guarantee. The guarantee is strictly positive, increasing in the number of bidders, and strictly below the interim benchmark of [Bergemann, Brooks and Morris \(2017\)](#) at every finite market size. Both converge to a common limit, and the coarse guarantee does so more slowly, at rate $\log N/N$ rather than $1/N$. When values differ across buyers, identical play need not be a worst case, but the coarse revenue guarantee is still bounded below by a system of equations similar to the one for common values, with one chain of constraints per buyer.

The worst case is sustained by states in which every bidder bids zero and the buyers receive the good at a price of zero, and the resulting surplus is what satisfies their obedience constraints. A reserve price reduces this surplus, but a small reserve

reduces it only slightly: coarse bidders pool at the reserve and pay it, the surplus falls only by the reserve, and the guarantee is continuous at a zero reserve. In the example of Section 6, the reserve that maximizes the guarantee is the smallest one at which the reserve pool vanishes, so that the worst case involves only opting out. Across the formats studied, the first-price auction has the highest guarantee: identical play is available at the same revenue in every standard auction, and where a deviator pays the others' bid rather than their own, the guarantee is zero. Finally, the guarantee is monotone in the convex order, so a seller who knows only the mean and an upper bound on the value should benchmark against the binary distribution.

For a seller who faces learning algorithms, the results suggest three things. The first-price auction is a natural choice, since it has the highest guarantee among the formats studied. A reserve price seems most useful when it is just large enough that opting out, rather than pooling at the reserve, becomes the worst case, since a small reserve is paid by the buyers and changes the guarantee little. When the value distribution is unknown, the binary distribution with the same mean is the appropriate benchmark.

Several questions remain. Away from common values the paper gives only a lower bound on the guarantee, and it is not known whether the bound is attained, or by which coupling of bids and values. Whether the first-price auction maximizes the coarse revenue guarantee over all standard auctions, as it maximizes the interim guarantee in [Bergemann, Brooks and Morris \(2019\)](#), is also open, since the ranking proved here covers only the formats in which a deviator who outbids tied opponents pays either their own bid or the opponents' bid. The optimal mechanism against coarsely rational bidders has not been characterized either, and the reserve price studied here is only one simple instrument. Finally, Remark 1 places the guarantee for contextual algorithms between the coarse and the interim guarantees, and characterizing it is left for future work.

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A Proofs for Sections 3 and 4

Ties

Every verification of a coarse constraint below checks it only at fixed bids δ that are not atoms of the law of the highest competing bid. Such δ form a dense set, since there are only countably many atoms, and at such δ ties have probability zero, so the deviation payoff is $\mathbb{E}[(v_i - \delta)\mathbb{1}\{M_i < \delta\}]$, where M_i is buyer i 's highest competing bid. For identical play, the following lemma extends the constraint from a dense set to every δ , ties included. For any joint law, it gives the closed form of the constraint, with $M_i \leq \delta$ in place of $M_i < \delta$, which every lower bound below uses. The only atoms checked directly are the bid of zero in the attainment step of Theorem 1 and the bid of exactly r under a reserve, where no admissible deviation lies below the atom.

Lemma 2 (Ties and the closed constraint). *Fix a reserve $r \in [0, \bar{v})$, with $r = 0$ meaning no reserve, a joint law of values and bids, and a buyer i with value v_i , expected payoff K_i , and highest competing bid M_i . Suppose that*

$$K_i \geq \mathbb{E}[(v_i - \delta)\mathbb{1}\{M_i < \delta\}] \quad (21)$$

for every δ in a dense subset of $(r, \bar{v}]$. Then (21) holds for every $\delta \in (r, \bar{v}]$, and

$$K_i \geq \mathbb{E}[(v_i - \delta)\mathbb{1}\{M_i \leq \delta\}] \quad \text{for every } \delta \in [r, \bar{v}]. \quad (22)$$

In a BCCE the hypothesis holds, since a deviation to a $\delta > r$ that is not an atom of the law of M_i ties with probability zero and wins exactly when $M_i < \delta$. If all buyers have the common value v and submit a common bid c , so that $M_i = c$, then every deviation to $\delta \in (r, \bar{v}]$ earns at most K_i , and without a reserve so does the deviation to 0.

Proof. For $\delta \in (r, \bar{v}]$ take $\delta_n \uparrow \delta$ in the dense set, and for $\delta \in [r, \bar{v})$ take $\delta_n \downarrow \delta$. The indicators $\mathbb{1}\{M_i < \delta_n\}$ converge to $\mathbb{1}\{M_i < \delta\}$ in the first case and to $\mathbb{1}\{M_i \leq \delta\}$ in the second, so dominated convergence gives (21) and (22) on these ranges. At $\delta = \bar{v}$, (22) follows from (21) because $v_i - \bar{v} \leq 0$. Under identical play, a deviator bidding $\delta \in (r, \bar{v}]$ wins outright when $c < \delta$ and shares with the $N - 1$ other buyers when $c = \delta$, so the deviation payoff is a convex combination of the right sides of (21) and

(22), hence at most K_i . Without a reserve, the deviation to 0 earns $\frac{1}{N}\mathbb{E}[v\mathbb{1}\{c=0\}]$, which is at most the right side of (22) at $\delta=0$ because $v \geq 0$. $\square$

Proof of Proposition 1

Throughout, for an action profile $a \in A$ let $a_{(1)} \geq a_{(2)} \geq \dots \geq a_{(N)}$ denote the order statistics and $M_i := \max_{j \neq i} a_j$ buyer i 's highest competing bid. For $x \in [0, \bar{v}]$ write $\vec{x} = (x, \dots, x)$ for the profile in which every buyer bids x , let $\mathbb{1}_{\vec{x}}$ denote the point mass at $\vec{x}$, and let $\otimes$ denote the product of measures.

The proposition follows from a single pooling step. There is no need to symmetrize σ first: under identical play every buyer faces the same coarse constraint, so it suffices that the constraints of σ hold on average across buyers.

Lemma 3 (Pooling). *Let $\sigma \in \mathcal{BCC}\mathcal{E}(\mu)$ and let γ be the image of σ under the Markov kernel*

$$\Gamma((v, a), \cdot) = \mathbb{1}_v \otimes \left[\frac{N-1}{N} \mathbb{1}_{\vec{a}_{(1)}} + \frac{1}{N} \mathbb{1}_{\vec{a}_{(2)}} \right].$$

Then $\gamma \in \mathcal{BCC}\mathcal{E}(\mu)$ is exchangeable identical play, and

$$R(\gamma) = \mathbb{E}_\sigma \left[\frac{N-1}{N} a_{(1)} + \frac{1}{N} a_{(2)} \right] \leq R(\sigma),$$

strictly unless $a_{(1)} = a_{(2)}$ holds σ -almost surely.

Proof. By construction, γ is an exchangeable identical play distribution with value marginal μ , and its revenue is the expected common bid. Since $a_{(2)} \leq a_{(1)}$ and $R(\sigma) = \mathbb{E}_\sigma[a_{(1)}]$, revenue weakly falls, with equality only if the top two bids are equal almost surely. Let J be uniform on $\{1, \dots, N\}$ and independent of (v, a) .

Step 1: the averaged law of the highest competing bid is unchanged. Condition on (v, a) . If $a_{(1)} > a_{(2)}$, exactly one buyer, the unique top bidder, faces $M_i = a_{(2)}$, and the remaining $N-1$ face $M_i = a_{(1)}$. If $a_{(1)} = a_{(2)}$, every buyer faces $M_i = a_{(1)}$. In both cases the conditional law of M_J is $\frac{N-1}{N} \mathbb{1}_{a_{(1)}} + \frac{1}{N} \mathbb{1}_{a_{(2)}}$, which by construction of Γ is the conditional law of the common bid c under γ . Hence the joint law of (v, M_J) , with (v, a) drawn from σ and J drawn independently, is the joint law of (v, c) under γ .

Step 2: deviation payoffs. Fix δ that is not an atom of the law of c . By Step 1, $\frac{1}{N} \sum_i \Pr(M_i = \delta) = \Pr(M_J = \delta) = 0$, so under σ a deviation to δ ties with probability

zero, and buyer i 's deviation payoff is $\mathbb{E}_\sigma[(v - \delta)\mathbb{1}\{M_i < \delta\}]$. Averaging over buyers and using Step 1,

$$\mathbb{E}_\gamma[(v - \delta)\mathbb{1}\{c < \delta\}] = \frac{1}{N} \sum_i \mathbb{E}_\sigma[(v - \delta)\mathbb{1}\{M_i < \delta\}] \leq \frac{1}{N} \sum_i \mathbb{E}_\sigma[u_i].$$

Step 3: equilibrium payoffs. The good is always allocated, so by (1) the buyers' total payoff is $v - a_{(1)}$ under σ and $v - c$ under γ , and under γ each buyer receives $1/N$ of the total. Hence

$$K_\gamma := \mathbb{E}_\gamma[u_i] = \frac{1}{N} \mathbb{E}_\sigma \left[v - \frac{N-1}{N} a_{(1)} - \frac{1}{N} a_{(2)} \right] \geq \frac{1}{N} \mathbb{E}_\sigma[v - a_{(1)}] = \frac{1}{N} \sum_i \mathbb{E}_\sigma[u_i].$$

Combining Steps 2 and 3, $K_\gamma \geq \mathbb{E}_\gamma[(v - \delta)\mathbb{1}\{c < \delta\}]$ for every δ that is not an atom of the law of c , so $\gamma \in \mathcal{BCC}\mathcal{E}(\mu)$ by Lemma 2. $\square$

Lemma 3 maps an arbitrary BCCE to an exchangeable identical play BCCE with weakly lower revenue, strictly unless the two highest bids of σ coincide almost surely, which proves the proposition. No minimizer need be assumed to exist. $\square$

B Proofs for Section 5

I prove Theorem 1 in its fixed-point form: the equation (10) has a unique solution $K^* \in (0, m)$, no identical play BCCE, and hence by Proposition 1 no BCCE at all, yields revenue below $m - NK^*$, and the comonotone coupling $c = \psi(F(v))$ attains it. The theorem follows by the substitution $q = F(v)$, under which the threshold is defined by $\Phi_\mu(F(\hat{v})) = K^*$, $\psi(F(v)) = H(v)$ as in (7), and (10) becomes the system of (5) and (6). The argument uses the density in Assumption 1 only to write the attaining bid as a function of the value, so it also proves Theorem 6.

Throughout, $v \sim \mu$, $R = \mathbb{E}[c]$, $K = (m - R)/N$, and $A(q) = \inf\{x : \Pr(c \leq x) \geq q\}$ is the quantile function of the bid, so $R = \int_0^1 A(q) dq$. Except where the density in Assumption 1 is invoked, μ is any non-degenerate prior on $[0, \bar{v}]$ and $F^{-1}(t) = \inf\{x : F(x) \geq t\}$ is its quantile function. Then Φ_μ is convex, nondecreasing, and absolutely continuous, with $\Phi_\mu(0) = 0$ and $\Phi_\mu(1) = m$, and for $K \in [0, m]$ the pooling quantile $q_K := \sup\{q \in [0, 1] : \Phi_\mu(q) \leq K\}$ satisfies $\Phi_\mu(q_K) = K$ and $\{q : \Phi_\mu(q) > K\} = (q_K, 1]$. Under Assumption 1, $q_K = \Phi_\mu^{-1}(K)$. One computation is

used repeatedly. Since Φ_μ is absolutely continuous with $\Phi'_\mu = F^{-1}$, for every $K \geq 0$ the map $p \mapsto (\Phi_\mu(p) - K)/p$ has derivative

$$\frac{F^{-1}(p)p - \Phi_\mu(p) + K}{p^2} = \frac{\int_0^p (F^{-1}(p) - F^{-1}(t))dt + K}{p^2} \geq 0 \quad (23)$$

almost everywhere on $(0, 1]$, so it is nondecreasing. The derivative is strictly positive when $K > 0$, and also when $K = 0$ under Assumption 1, since F^{-1} is then strictly increasing.

Lemma 4 (Rearrangement bound). *For any event B with $\Pr(B) = q$, $\mathbb{E}[v\mathbb{1}_B] \geq \Phi_\mu(q)$.*

Proof. This is the Hardy–Littlewood rearrangement inequality (Hardy, Littlewood and Pólya, 1934): among events of probability q , the expectation $\mathbb{E}[v\mathbb{1}_B]$ is smallest when B is placed on the lowest realizations of v , splitting an atom of μ if necessary, which gives $\int_0^q F^{-1}(t)dt$. $\square$

Lemma 4 lets me derive the monotonicity of the worst-case bidding rule rather than assume it. The next two lemmas are the two halves of the argument: a lower bound and an attainment check.

Lemma 5 (Chain). *Let X be a random variable with values in $[0, \bar{v}]$ on the same probability space as $v \sim \mu$, with cdf F_X and quantile function $A(q) = \inf\{x : F_X(x) \geq q\}$. Let $K \geq 0$ and $\Delta \subseteq [0, \bar{v}]$, and suppose*

$$K \geq \mathbb{E}[(v - \delta)\mathbb{1}\{X \leq \delta\}] \quad \text{for every } \delta \in \Delta.$$

Then $A(q) \geq (\Phi_\mu(q) - K)/q$ for every $q \in (0, 1)$ with $A(q) \in \Delta$. In particular, if $\Delta = [0, \bar{v}]$, then $\mathbb{E}[X] \geq \mathcal{R}_\mu(K)$.

Proof. By Lemma 4, the hypothesis gives $K \geq \Phi_\mu(F_X(\delta)) - \delta F_X(\delta)$ for every $\delta \in \Delta$. Fix q with $\delta = A(q) \in \Delta$. Right-continuity of F_X gives $F_X(A(q)) \geq q$, and the map $p \mapsto (\Phi_\mu(p) - K)/p$ is nondecreasing by (23). Hence $A(q) \geq (\Phi_\mu(F_X(A(q))) - K)/F_X(A(q)) \geq (\Phi_\mu(q) - K)/q$. If $\Delta = [0, \bar{v}]$ this holds for every q , and since $A \geq 0$, $\mathbb{E}[X] = \int_0^1 A(q) dq \geq \int_0^1 (\Phi_\mu(q) - K)^+/q dq = \mathcal{R}_\mu(K)$. $\square$

Lemma 6 (Binding bids). *Let U be uniform on $[0, 1]$, $v = F^{-1}(U)$, and $c = C(U)$, where $C : [0, 1] \rightarrow [0, \bar{v}]$ is nondecreasing and, for some $q_0 \in [0, 1)$ and $K \geq 0$,*

satisfies $C(q) = (\Phi_\mu(q) - K)/q$ on $(q_0, 1]$. If $K > 0$ or Assumption 1 holds, then C is continuous and strictly increasing on $(q_0, 1]$, and the identical play distribution with common bid c satisfies

$$\mathbb{E}\left[(v - \delta)\left(\mathbb{1}\{c < \delta\} + \frac{1}{N}\mathbb{1}\{c = \delta\}\right)\right] \leq K \quad \text{for every } \delta > C(q_0^+),$$

with equality for $\delta \in (C(q_0^+), C(1)]$, where $C(q_0^+) = \lim_{q \downarrow q_0} C(q)$.

Proof. By (23), C is continuous and strictly increasing on $(q_0, 1]$, with $C(1) = m - K$. Fix $\delta \in (C(q_0^+), C(1)]$ and let $q \in (q_0, 1]$ be the unique solution of $C(q) = \delta$. Since C is nondecreasing and $C(q_0^+) < \delta$, $\{c < \delta\} = \{U < q\}$ up to a null set and $\Pr(c = \delta) = \Pr(U = q) = 0$, so the deviation payoff is $\mathbb{E}[(F^{-1}(U) - \delta)\mathbb{1}\{U < q\}] = \Phi_\mu(q) - \delta q = K$. For $\delta > C(1)$ the deviator always wins and earns $m - \delta < K$. $\square$

Step 1: the lower bound. Fix any identical play $\gamma \in \mathcal{BCC}\mathcal{E}(\mu)$. By Lemma 2 without a reserve and with $M_i = c$,

$$K \geq \mathbb{E}[(v - \delta)\mathbb{1}\{c \leq \delta\}] \quad \text{for every } \delta \in [0, \bar{v}]. \quad (24)$$

Lemma 5 with $X = c$ and $\Delta = [0, \bar{v}]$ then gives

$$R = \mathbb{E}[c] \geq \mathcal{R}_\mu(K). \quad (25)$$

Together with $R = m - NK$ this gives $g_N(K) \geq 0$, where $g_N(K) := m - NK - \mathcal{R}_\mu(K)$.

Lemma 7 (Concavity and uniqueness). $\mathcal{R}_\mu$ is continuous, convex, and strictly decreasing on $[0, m]$, with $\mathcal{R}_\mu(m) = 0$. Hence g_N is concave on $[0, m]$, with $g_N(0) = m - \int_0^1 \frac{\Phi_\mu(q)}{q} dq > 0$ and $g_N(m) = -(N-1)m < 0$, so g_N has a unique zero $K^* \in (0, m)$, is positive on $[0, K^*)$ and negative on $(K^*, m]$.

Proof. For each q , the map $K \mapsto (\Phi_\mu(q) - K)^+/q$ is convex and nonincreasing and is bounded by $\Phi_\mu(q)/q \leq F^{-1}(q)$. Hence $\mathcal{R}_\mu$ is convex and nonincreasing, and continuous by dominated convergence. It is strictly decreasing because for $K < m$ the integrand strictly decreases in K on $(q_K, 1]$, a set of positive measure, and $\mathcal{R}_\mu(m) = 0$ because $\Phi_\mu \leq m$. For the left endpoint, $\Phi_\mu(q)/q$ is the average of the nondecreasing function F^{-1} over $[0, q]$, which is strictly below its average m over $[0, 1]$ for every $q < 1$ because μ is non-degenerate, so $g_N(0) > 0$. A concave function positive at the left

endpoint and negative at the right endpoint has exactly one zero and crosses from above. $\square$

By Lemma 7, $g_N(K) \geq 0$ forces $K \leq K^*$ and therefore $R = m - NK \geq m - NK^*$, which is the required lower bound.

Step 2: attainment. Let U be uniform on $[0, 1]$ and consider the identical play distribution with value $v = F^{-1}(U)$ and common bid $c = \psi(U)$, which is $C(U)$ with $C = \psi$ and $q_0 = q_{K^*}$. Under Assumption 1, $U = F(v)$ almost surely, so this is the comonotone coupling $c = \psi(F(v))$. The value marginal is μ , revenue is $\int_0^1 \psi = \mathcal{R}_\mu(K^*) = m - NK^*$, and the rent is K^* . Since $\Phi_\mu(q_{K^*}) = K^*$, we have $\psi(q_{K^*}^+) = 0$, so Lemma 6, which applies because $K^* > 0$, covers every deviation $\delta > 0$, and the constraint binds for $\delta \in (0, m - K^*]$. At $\delta = 0$ the deviator ties with the atom at zero and wins with probability $1/N$, earning $\frac{1}{N}\mathbb{E}[v\mathbb{1}\{c = 0\}] = \Phi_\mu(q_{K^*})/N = K^*/N < K^*$. Hence the coupling is a BCCE, and $CRG(\mu) = m - NK^* = \mathcal{R}_\mu(K^*)$, which is strictly positive because $K^* < m$. Moreover, $\psi(q) \leq F^{-1}(q)$ pointwise, since $\Phi_\mu(q) \leq qF^{-1}(q)$: the worst-case bid never exceeds the value, as required, since a bidder pooled above the value could profitably commit to zero. $\square$

Proof of Proposition 2. With $g_N(K) = m - NK - \mathcal{R}_\mu(K)$, Lemma 7 gives a unique root crossing from above. Since $g_{N+1} < g_N$ pointwise on $(0, m]$, the root falls, $K_{N+1}^* < K_N^*$. As $\mathcal{R}_\mu$ is strictly decreasing on $[0, m)$ and $CRG = \mathcal{R}_\mu(K_N^*)$, the guarantee rises. $\square$

The strict coarseness gap

Proof of Proposition 4. Let σ_B be a BCE attaining $RG_N(\mu)$ whose two highest bids differ with positive probability. Since $\mathcal{BCE} \subseteq \mathcal{BCCCE}$, σ_B is a BCCE with revenue $RG_N(\mu)$, and Proposition 1 yields an identical play BCCE with revenue strictly below $RG_N(\mu)$. Hence $CRG_N(\mu) < RG_N(\mu)$. $\square$

Asymptotic convergence

Proof of Theorem 2. From $CRG_N = m - NK_N^* \geq 0$ we get $K_N^* \leq m/N \rightarrow 0$. The integrand in (9) satisfies $0 \leq \frac{(\Phi_\mu(q) - K)^+}{q} \leq \frac{\Phi_\mu(q)}{q} \leq \bar{v}$ and converges pointwise to $\frac{\Phi_\mu(q)}{q}$ as $K \downarrow 0$ whenever $\Phi_\mu(q) > 0$, which holds for every $q > 0$ under Assumption

1. Dominated convergence gives $CRG_N = \mathcal{R}_\mu(K_N^*) \rightarrow \int_0^1 \frac{\Phi_\mu(q)}{q} dq$, and substituting $q = F(v)$ turns this into $\int_0^{\bar{v}} G(v) f(v) dv$, the limiting revenue guarantee of [Bergemann, Brooks and Morris \(2017\)](#). $\square$

C Reserve Prices

Proof of Proposition 5

For $r = 0$ the kernel Γ_r below is that of Lemma 3 and the claim is Proposition 1, so let $r > 0$. With a reserve price r , bidding in $[0, r)$ is equivalent to bidding 0, and I use this normalization throughout. Fix a BCCE σ of the auction with reserve r , write $\bar{K}_\sigma := \frac{1}{N} \sum_i \mathbb{E}_\sigma[u_i]$ for the buyers' average payoff, and let J be uniform on $\{1, \dots, N\}$ and independent of (v, a) . As in Lemma 3, no symmetrization is needed.

Write $b := \max\{a_{(2)}, r\}$ and define the Markov kernel Γ_r on $V \times A$ by

$$\Gamma_r((v, a), \cdot) = \begin{cases} \mathbb{1}_v \otimes \mathbb{1}_{\vec{0}}, & a_{(1)} < r, \\ \mathbb{1}_v \otimes \left[\frac{N-1}{N} \mathbb{1}_{\vec{a}_{(1)}} + \frac{1}{N} \mathbb{1}_{\vec{b}} \right], & a_{(1)} \geq r \text{ and } (a_{(2)} \geq r \text{ or } v \geq a_{(1)}), \\ \mathbb{1}_v \otimes \left[\frac{N-1}{N} \mathbb{1}_{\vec{a}_{(1)}} + \frac{1}{N} \mathbb{1}_{\vec{0}} \right], & a_{(1)} \geq r > a_{(2)} \text{ and } v < a_{(1)}, \end{cases}$$

and let γ be the image of $\sigma \Gamma_r$ under one further normalization: on the event where the common bid equals r and $v < r$, replace the common bid by $\vec{0}$. The resulting γ is an exchangeable identical play distribution with value marginal μ , and by construction $v \geq r$ almost surely on the event that the common bid equals r . Write c for the common bid under γ . The proposition then follows from three claims.

Claim 1: revenue weakly falls. On $\{a_{(1)} < r\}$ both revenues are zero. On the second branch, revenue changes from $a_{(1)}$ to $\frac{N-1}{N} a_{(1)} + \frac{1}{N} \max\{a_{(2)}, r\} \leq a_{(1)}$, since $a_{(2)} \leq a_{(1)}$ and $r \leq a_{(1)}$. On the third branch it changes from $a_{(1)}$ to $\frac{N-1}{N} a_{(1)} \leq a_{(1)}$. The final normalization lowers revenue further, from r to 0 on the states it touches.

Claim 2: the buyers' total payoff weakly rises. By (13) the total payoff is $(v - a_{(1)}) \mathbb{1}\{a_{(1)} \geq r\}$. On $\{a_{(1)} < r\}$ it is zero before and after. On the second branch it changes from $v - a_{(1)}$ to $v - \left[\frac{N-1}{N} a_{(1)} + \frac{1}{N} \max\{a_{(2)}, r\} \right] \geq v - a_{(1)}$. On the third branch, where $v < a_{(1)}$ so the sale yields negative surplus, it changes from $v - a_{(1)} < 0$ to $\frac{N-1}{N} (v - a_{(1)})$, which is strictly larger. The final normalization raises the total further on the states it changes, from the negative surplus $v - r < 0$ to 0. Under γ

each buyer receives $1/N$ of the total, so $K_\gamma \geq \bar{K}_\sigma$.

Claim 3: deviation payoffs weakly fall. A deviation to $\delta < r$ never wins under either distribution and earns zero, while $K_\gamma \geq \bar{K}_\sigma \geq 0$ because deviating to 0 is available. Fix $\delta > r$ that is not an atom of the law of c . An atom of the law of M_J at δ requires an atom of $a_{(1)}$ or $a_{(2)}$ there, which the kernel maps to an atom of c at the same point, so δ is not an atom of the law of M_J , nor of the law of any M_i . Couple M_J and c through the randomization in the kernel, as in Step 1 of Lemma 3. On the second branch with $a_{(2)} \geq r$ the kernel is that of Lemma 3, and Step 1 of that lemma gives the equality of the conditional laws of M_J and c . On every other branch, and after the final normalization, whenever the coupled pair (M_J, c) differs both coordinates are at most $r < \delta$, so both indicators equal one. Hence, for every such δ ,

$$\mathbb{E}_\gamma[(v - \delta)\mathbb{1}\{c < \delta\}] = \frac{1}{N} \sum_i \mathbb{E}_\sigma[(v - \delta)\mathbb{1}\{M_i < \delta\}] \leq \bar{K}_\sigma \leq K_\gamma.$$

By Lemma 2, every deviation to $\delta \in (r, \bar{v}]$ earns at most K_γ , and $K_\gamma \geq B := \mathbb{E}_\gamma[(v - r)\mathbb{1}\{c \leq r\}]$. The deviation to exactly r is the one level left, since there are no admissible deviations below the reserve from which to take the upward limit. A deviator bidding r wins when $c = 0$ and ties the full pool when $c = r$, so the deviation payoff is $(1 - \frac{1}{N})A + \frac{1}{N}B$ with $A = \mathbb{E}_\gamma[(v - r)\mathbb{1}\{c = 0\}]$. The normalization placed $v \geq r$ on $\{c = r\}$, so $B - A = \mathbb{E}_\gamma[(v - r)\mathbb{1}\{c = r\}] \geq 0$ and the deviation payoff is at most $B \leq K_\gamma$. Hence γ is a BCCE. $\square$

The two-pool family is feasible

Lemma 8 (Feasibility). *Fix r , and let (s, q_0) satisfy (14) and (15) with $K = \Phi_\mu(q_0) - rq_0 \geq 0$ and $\Phi_\mu(q_0) - \Phi_\mu(s) \geq r(q_0 - s)$. Then the comonotone identical play distribution with common bid $c = 0$ on the lowest s quantiles of v , $c = r$ on quantiles $[s, q_0)$, and $c = \psi_r(q) := (\Phi_\mu(q) - K)/q$ at quantile $q \geq q_0$ is a BCCE of the auction with reserve r , with revenue R as in (15).*

Proof. The construction has value marginal μ , and by (13) and (15) each buyer's payoff is exactly K . The common bid is $C(U)$ with $U = F(v)$, where $C = 0$ on $[0, s)$, $C = r$ on $[s, q_0)$, and $C = \psi_r$ on $[q_0, 1]$, with $\psi_r(q_0) = r$ by (14). Lemma 6 covers every deviation $\delta > r$. For $\delta \in (0, r)$ the deviator never clears the reserve and earns

$0 \leq K$. For $\delta = 0$ the deviator ties the opt-out pool below the reserve, no sale occurs, and the payoff is again $0 \leq K$. For $\delta = r$ the deviator wins the opt-out pool and ties the reserve pool, earning

$$\begin{aligned} (\Phi_\mu(s) - rs) + \frac{1}{N}(\Phi_\mu(q_0) - \Phi_\mu(s) - r(q_0 - s)) \\ = K - (1 - \frac{1}{N})(\Phi_\mu(q_0) - \Phi_\mu(s) - r(q_0 - s)) \leq K, \end{aligned}$$

where the identity substitutes (14) and the inequality is the feasibility condition. Hence the construction is a BCCE. $\square$

Proof of Theorem 3: exactness

For $r \geq m$, the distribution in which every buyer always bids 0 is a BCCE, since a deviation to any $\delta \geq r$ wins and earns $m - \delta \leq 0$, so the guarantee is zero. For $r = 0$ the integrand in (16) is $(\Phi_\mu(q) - K)^+/q$. A feasible share has $\Phi_\mu(s) \leq \kappa_+(s)$, so this integrand vanishes on $[0, s]$ at $K = \kappa_+(s)$, and (16) reads $g_N(\kappa_+(s)) = \Phi_\mu(s) \geq 0$. Hence $\kappa_+(s) \leq K^*$ by Lemma 7, and the revenue $m - \Phi_\mu(s) - N\kappa_+(s) = \mathcal{R}_\mu(\kappa_+(s))$ is at least $\mathcal{R}_\mu(K^*) = CRG_N(0)$, with equality at $s = 0$, where $\kappa_+(0) = K^*$. Hence $\widehat{V}_N(0) = CRG_N(0)$, which is Theorem 1. Fix now $r \in (0, m)$.

Definitions. The function $q \mapsto \Phi_\mu(q) - rq$ vanishes at 0, decreases on $[0, F(r)]$ and increases thereafter, so for $\kappa \geq 0$ its sublevel set $\{q : \Phi_\mu(q) - rq \leq \kappa\}$ is an interval $[0, q_r(\kappa)]$ with q_r nondecreasing, and $(\Phi_\mu(q) - \kappa)/q > r$ exactly for $q > q_r(\kappa)$. Write $\underline{C}_\kappa(q) := \max\{r, (\Phi_\mu(q) - \kappa)/q\}$, which is nonincreasing in κ , and for $s \in [0, 1]$ and $\kappa \in [0, m]$

$$h_s(\kappa) := m - \Phi_\mu(s) - N\kappa - \int_s^1 \underline{C}_\kappa(q) dq,$$

so that (16) reads $h_s(K) = 0$. The function h is jointly continuous in (s, κ) , and $h_s(m) \leq m - Nm < 0$. Whenever $\{\kappa \in [0, m] : h_s(\kappa) \geq 0\}$ is nonempty, its supremum is the largest solution $\kappa_+(s)$ of (16), with $\kappa_+(s) < m$ and $h_s(\kappa_+(s)) = 0$. Set $\mathcal{F}(s) := m - \Phi_\mu(s) - N\kappa_+(s)$, and call s *admissible* if $\kappa_+(s)$ is defined and $\Phi_\mu(s) - rs \leq \kappa_+(s)$, which is feasibility in the sense of Section 6. The theorem therefore claims that $\mathcal{F}$ attains a minimum over the admissible shares and that this minimum is $CRG_N(r)$.

Step 1: the lower bound. Fix an identical play BCCE with common bid $c \in \{0\} \cup [r, \bar{v}]$, rent K , revenue R , opt-out share $s := \Pr(c = 0)$, and bid quantile

function C . Two bounds hold for every coupling of c with v . First, by Lemma 2 the closed constraint $K \geq \mathbb{E}[(v - \delta)\mathbb{1}\{c \leq \delta\}]$ holds for every $\delta \in [r, \bar{v}]$, so Lemma 5 with $X = c$ and $\Delta = [r, \bar{v}]$ gives $C(q) \geq (\Phi_\mu(q) - K)/q$ whenever $C(q) \geq r$, that is for $q > s$, and hence

$$R = \int_s^1 C(q) dq \geq \int_s^1 \underline{C}_K(q) dq.$$

Second, (13) gives $NK = m - \mathbb{E}[v\mathbb{1}\{c = 0\}] - R \leq m - \Phi_\mu(s) - R$ by Lemma 4. Together, $h_s(K) \geq 0$, so $\kappa_+(s)$ is defined and $K \leq \kappa_+(s)$, and since $\underline{C}_\kappa$ is nonincreasing in κ ,

$$R \geq \int_s^1 \underline{C}_{\kappa_+(s)}(q) dq = \mathcal{F}(s),$$

where the equality is $h_s(\kappa_+(s)) = 0$. Moreover, $\Phi_\mu(s) - rs \leq K \leq \kappa_+(s)$, so the share s is admissible. Otherwise $s > 0$, and by continuity there is $\eta > 0$ with $(\Phi_\mu(q) - K)/q > r$ on $(s, s + \eta)$, so the bound on C forces $C > r$ there, there is no bid atom at r , and $F_c(r) = s$. The closed constraint at $\delta = r$ with Lemma 4 then reads $K \geq \Phi_\mu(s) - rs$, a contradiction. By Proposition 5, $CRG_N(r) \geq \inf\{\mathcal{F}(s) : s \text{ admissible}\}$.

Step 2: the candidate for an admissible share. First, $\kappa_+(s) < m - r$ for every admissible s . At $\kappa \geq m - r$ the envelope $\underline{C}_\kappa$ equals r throughout $(s, 1]$, since $(\Phi_\mu(q) - \kappa)/q \leq m - \kappa \leq r$, so h_s is affine with slope $-N$ on $[m - r, m]$, and $h_s(m - r) = rs - \Phi_\mu(s) - (N - 1)(m - r) \leq rF(r) - \Phi_\mu(F(r)) - (N - 1)(m - r) < 0$ by the hypothesis of the theorem, since $s \mapsto rs - \Phi_\mu(s)$ is maximized where $F^{-1}(s) = r$. Hence the crossing $q_0 := q_r(\kappa_+(s))$ is interior and, by continuity, $\Phi_\mu(q_0) - rq_0 = \kappa_+(s)$. For admissible s , consider the distribution with bids 0 on the lowest s quantiles and $\underline{C}_{\kappa_+(s)}$ above. Admissibility says $\Phi_\mu(s) - rs \leq \kappa_+(s)$, equivalently $q_0 \geq s$, so the bids equal r on (s, q_0) and $(\Phi_\mu(q) - \kappa_+(s))/q$ above. The identity $\Phi_\mu(q_0) - rq_0 = \kappa_+(s)$ is (14), and $h_s(\kappa_+(s)) = 0$ is (15), so the candidate is a member of the two-pool family with revenue $\mathcal{F}(s)$, and admissibility is its feasibility condition $\Phi_\mu(q_0) - \Phi_\mu(s) \geq r(q_0 - s)$.

Step 3: attainment. For admissible s , the candidate of Step 2 satisfies the hypotheses of Lemma 8, so it is a BCCE with revenue $\mathcal{F}(s)$. With Step 1, $CRG_N(r) = \inf\{\mathcal{F}(s) : s \text{ admissible}\}$, and the admissible set is nonempty by Step 1 applied to any BCCE, for instance the one in which every buyer bids m . It remains to show that the infimum is attained. Let admissible shares s_n converge to s , and pass to a subsequence along which $\kappa_+(s_n)$ converges to some κ . By joint continuity, $h_s(\kappa) = \lim_n h_{s_n}(\kappa_+(s_n)) = 0$, so $\kappa_+(s)$ is defined and at least κ , and

$\Phi_\mu(s) - rs \leq \kappa \leq \kappa_+(s)$. Hence the admissible set is closed and κ_+ is upper semicontinuous on it, so $\mathcal{F}$ is lower semicontinuous there. The share $s = 1$ is not admissible, since $\kappa_+(1) = 0 < m - r = \Phi_\mu(1) - r$, so the admissible set is a compact subset of $[0, 1)$ and $\mathcal{F}$ attains its minimum on it. Therefore $CRG_N(r) = \widehat{V}_N(r)$. $\square$

D Symmetric and Asymmetric Priors

Throughout, π is a prior on V^N with per-buyer marginals $\mu_1, \dots, \mu_N$, a BCCE is a joint law σ of $(v_1, \dots, v_N, a_1, \dots, a_N)$ with value marginal π satisfying the coarse constraints, $u_i = q_i(a)(v_i - a_i)$, $M_i := \max_{j \neq i} a_j$, and $K_i := \mathbb{E}_\sigma[u_i]$. Since the good is always allocated, $\sum_i u_i = v_{\text{win}} - a_{(1)}$, where v_{win} is the winner's value.

Symmetrization. For exchangeable π , averaging a BCCE σ over its $N!$ images under $(v, a) \mapsto (v_\rho, a_\rho)$, with $(v_\rho)_i = v_{\rho(i)}$ and $(a_\rho)_i = a_{\rho(i)}$ for a bijection ρ of $\{1, \dots, N\}$, gives an exchangeable BCCE with the same revenue and with $K_i = K$ for every i : the value marginal of each image is π by exchangeability of π , $u_i(v_\rho, a_\rho) = u_{\rho(i)}(v, a)$ and $M_i(a_\rho) = M_{\rho(i)}(a)$ make each image a BCCE, the constraint set is convex, and $a_{(1)}$ is permutation invariant.

Proof of Theorem 5

The set of $R \in [0, m_N]$ satisfying (19) is closed, since each $\mathcal{R}_{\mu_i}^{-1}$ is continuous by Lemma 7 applied to μ_i , and it is nonempty, since at $R = \max_i \mathcal{R}_{\mu_i}(0)$ every term on the left is zero while $\mathcal{R}_{\mu_i}(0) < m_i \leq m_N$. Hence $\tilde{R}$ is well defined.

Fix a BCCE σ with revenue $R = \mathbb{E}[a_{(1)}]$ and rents $K_1, \dots, K_N$. For each buyer i , Lemma 2 gives $K_i \geq \mathbb{E}[(v_i - \delta)\mathbb{1}\{M_i \leq \delta\}]$ for every $\delta \in [0, \bar{v}]$, and Lemma 5 applied to the pair (v_i, M_i) , which uses only the marginal $v_i \sim \mu_i$, gives

$$\mathbb{E}[M_i] \geq \mathcal{R}_{\mu_i}(K_i) \quad \text{for every buyer } i. \quad (26)$$

Since $M_i \leq a_{(1)}$ pointwise, $R \geq \mathcal{R}_{\mu_i}(K_i)$, and since $\mathcal{R}_{\mu_i}$ is nonincreasing this forces $K_i \geq \mathcal{R}_{\mu_i}^{-1}(R)$, for otherwise $\mathcal{R}_{\mu_i}(K_i) > R$. Since the winner's value is at most the highest value, $\sum_i K_i = \mathbb{E}[v_{\text{win}} - a_{(1)}] \leq m_N - R$. Summing, $\sum_i \mathcal{R}_{\mu_i}^{-1}(R) \leq \sum_i K_i \leq m_N - R$, so R satisfies (19) and $R \geq \tilde{R}$. Taking the infimum over BCCE gives the inequality. $\square$

Proof of Theorem 4

Let $\tilde{g}(K) := m_N - NK - \mathcal{R}_\mu(K) = g_N(K) + (m_N - m)$. By Lemma 7, $\tilde{g}$ is concave on $[0, m]$ with $\tilde{g}(0) \geq g_N(0) > 0$, and $\tilde{g}(m) = m_N - Nm \leq 0$ since $\mathcal{R}_\mu(m) = 0$ and $\max_i v_i \leq \sum_i v_i$. Hence $\tilde{g}$ has a unique zero $\tilde{K} \in (0, m]$, is positive on $[0, \tilde{K})$ and negative on $(\tilde{K}, m]$, and $\tilde{K} \geq K^*$ with strict inequality if and only if $m_N > m$.

Theorem 5 applies with $\mu_i = \mu$ for every i , so it remains to show that $\tilde{R} = m_N - N\tilde{K}$. For $R \in [0, m_N]$ write $K(R) := (m_N - R)/N \in [0, m]$. Since $\mathcal{R}_\mu$ is continuous and nonincreasing, $\{K : \mathcal{R}_\mu(K) \leq R\} = [\mathcal{R}_\mu^{-1}(R), m]$, so (19) holds if and only if $\mathcal{R}_\mu(K(R)) \leq R$, that is $\tilde{g}(K(R)) \geq 0$, that is $K(R) \leq \tilde{K}$, that is $R \geq m_N - N\tilde{K}$. Hence $\tilde{R} = m_N - N\tilde{K} = \mathcal{R}_\mu(\tilde{K})$, and Theorem 5 gives $CRG(\pi) \geq m_N - N\tilde{K}$. $\square$

Supplementary Appendix

This supplement collects the proofs of the rate of convergence (Proposition 3), the continuity of the guarantee at a zero reserve (Theorem 3), and the results of Section 9. It also records the slope of the guarantee at a zero reserve (Proposition S.1). Notation is that of the main appendix.

S.1 Rate of Convergence

Lemma 9 (The interim guarantee converges at rate $1/N$). *For every $N \geq 2$, $0 \leq R_\infty - RG_N \leq \frac{2\bar{v}}{N-1}$.*

Proof. Write $\alpha = \frac{N-1}{N}$. Substituting $t = uF(v)$ in the two conditional means gives

$$G(v) = \int_0^1 F^{-1}(uF(v)) du, \quad \beta_N(v) = \int_0^1 F^{-1}(uF(v)) \alpha u^{\alpha-1} du,$$

so $G(v) - \beta_N(v) = \int_0^1 F^{-1}(uF(v)) (1 - \alpha u^{\alpha-1}) du$. The weight $1 - \alpha u^{\alpha-1}$ integrates to zero, is negative for $u < u_0 := (1 - \frac{1}{N})^N$ and positive above, and $F^{-1}(uF(v))$ is nondecreasing in u , so the integral is nonnegative by the Chebyshev integral inequality: $\beta_N \leq G$ pointwise. For the upper bound, $|G - \beta_N| \leq \bar{v} \int_0^1 |1 - \alpha u^{\alpha-1}| du = 2\bar{v} (u_0^\alpha - u_0) = 2\bar{v} \frac{u_0}{N-1} \leq \frac{2\bar{v}}{N-1}$, using $u_0^{\alpha-1} = (1 - \frac{1}{N})^{-1}$. Integrating against f gives the claim. $\square$

Proof of Proposition 3. The bound $0 \leq R_\infty - RG_N \leq 2\bar{v}/(N-1)$ is Lemma 9. Write $K_N := K_N^*$ for the rent at market size N and $q_N := \Phi_\mu^{-1}(K_N)$ for the pooling quantile, so that $\Phi_\mu(q_N) = K_N$. The proof of Theorem 2 gives $R_\infty = \int_0^1 \frac{\Phi_\mu(q)}{q} dq$ and $K_N \leq m/N \rightarrow 0$, hence $q_N \rightarrow 0$.

Step 1: an exact decomposition. Since $\Phi_\mu(q) \geq K_N$ exactly when $q \geq q_N$, (9) reads $CRG_N = \int_{q_N}^1 \frac{\Phi_\mu(q) - K_N}{q} dq$, and subtracting from R_∞ ,

$$R_\infty - CRG_N = T_N + K_N \log \frac{1}{q_N}, \quad T_N := \int_0^{q_N} \frac{\Phi_\mu(q)}{q} dq.$$

The first term is the value on the pooled quantiles, and the second is the rent on the bidding quantiles.

Step 2: the order of the rent. By (10), $NK_N = m - CRG_N \rightarrow m - R_\infty$, and this limit is strictly positive because $\Phi_\mu(q)/q$, the average of the strictly increasing function F^{-1} over $[0, q]$, is strictly below $F^{-1}(q)$, so $R_\infty < \int_0^1 F^{-1} = m$. Hence $K_N = \frac{m-R_\infty}{N}(1 + o(1))$ and $\log \frac{1}{K_N} = \log N + O(1)$.

Step 3: the order of the pooling quantile. This is the only step that uses the continuity and positivity of f at 0 in Assumption 1. They give $F(x) = f(0)x(1 + o(1))$ as $x \downarrow 0$, hence $F^{-1}(t) = \frac{t}{f(0)}(1 + o(1))$ as $t \downarrow 0$ and, integrating, $\Phi_\mu(q) = \frac{q^2}{2f(0)}(1 + o(1))$ as $q \downarrow 0$. At $q = q_N$ this reads $K_N = \frac{q_N^2}{2f(0)}(1 + o(1))$, so

$$\log \frac{1}{q_N} = \frac{1}{2} \log \frac{1}{K_N} + O(1) = \frac{1}{2} \log N + O(1).$$

The square root in $q_N \asymp \sqrt{K_N}$ is the source of the factor $\frac{1}{2}$ in the proposition.

Step 4: the pooled term is of smaller order. Since $\Phi_\mu(q)/q \leq F^{-1}(q)$, $0 \leq T_N \leq \int_0^{q_N} F^{-1}(q) dq = \Phi_\mu(q_N) = K_N = O(1/N)$.

Step 5: conclusion. By Steps 2 and 3, $K_N \log \frac{1}{q_N} = \frac{m-R_\infty}{N}(1 + o(1))(\frac{1}{2} \log N + O(1)) = \frac{(m-R_\infty) \log N}{2N}(1 + o(1))$, and adding $T_N = O(1/N) = o(\log N/N)$ gives the expansion for $R_\infty - CRG_N$. Finally $RG_N - CRG_N = (R_\infty - CRG_N) - (R_\infty - RG_N)$, and the second bracket is $O(1/N) = o(\log N/N)$ by Lemma 9, so $RG_N - CRG_N$ has the same expansion. $\square$

S.2 Small Reserves

Proof of Theorem 3: continuity at a zero reserve

Let K^* be the rent and $q^* := q_{K^*}$ the pooling quantile at $r = 0$, so that $CRG_N(0) = m - NK^* = \mathcal{R}_\mu(K^*)$ by Theorem 1, and recall that $g_N(0) > 0$. I show that for every $r \in (0, g_N(0))$,

$$CRG_N(0) - r \log \frac{1}{q^*} \leq CRG_N(r) \leq CRG_N(0) + r \frac{NK^*}{g_N(0)}, \quad (27)$$

which gives $\lim_{r \downarrow 0} CRG_N(r) = CRG_N(0)$. Neither bound uses the hypothesis of the exactness part of Theorem 3. The notation is that of Appendix C.

Lower bound. By Proposition 5 it suffices to bound the revenue R of an identical play BCCE with common bid $c \in \{0\} \cup [r, \bar{v}]$, rent K , and opt-out share s . Step

1 of the exactness proof in Appendix C gives $R \geq \int_s^1 \underline{C}_K \geq \int_s^1 (\Phi_\mu(q) - K)^+/q \, dq$, $NK \leq m - \Phi_\mu(s) - R$, and $\Phi_\mu(s) - rs \leq K$. The first two inequalities give

$$NK \leq m - \mathcal{R}_\mu(K) - \left[\Phi_\mu(s) - \int_0^s \frac{(\Phi_\mu(q) - K)^+}{q} \, dq \right] \leq m - \mathcal{R}_\mu(K),$$

where the bracket is nonnegative because $(\Phi_\mu(q) - K)^+/q \leq \Phi_\mu(q)/q \leq F^{-1}(q)$. Hence $g_N(K) \geq 0$, so $K \leq K^*$ by Lemma 7, and

$$R \geq \int_s^1 \frac{(\Phi_\mu(q) - K^*)^+}{q} \, dq = CRG_N(0) - \int_0^s \frac{(\Phi_\mu(q) - K^*)^+}{q} \, dq.$$

The last integrand vanishes for $q \leq q^*$, and for $q \in (q^*, s]$ it is at most $(\Phi_\mu(s) - K)/q \leq rs/q$. The integral is therefore zero if $s \leq q^*$, and at most $rs \log(s/q^*) \leq r \log(1/q^*)$ otherwise.

Upper bound. Let $\kappa_r := K^*(1 - r/g_N(0)) \in (0, K^*)$. Since g_N is concave with $g_N(K^*) = 0$, $g_N(\kappa_r) \geq (r/g_N(0))g_N(0) = r$. Since $\max\{r, x\} \leq x^+ + r$, $h_0(\kappa) \geq g_N(\kappa) - r$ for every κ , so $h_0(\kappa_r) \geq 0$ and $\kappa_+(0) \geq \kappa_r$, and the share $s = 0$ is admissible. On $[m - r, m]$ the function h_0 is affine with slope $-N$ and $h_0(m - r) = -(N - 1)(m - r) < 0$, so $\kappa_+(0) < m - r$ and $q_0 := q_r(\kappa_+(0)) < 1$. As in Step 2 of the exactness proof, the candidate with $s = 0$ and pool boundary q_0 is a member of the two-pool family, and by Lemma 8 it is a BCCE with revenue $\mathcal{F}(0) = m - N\kappa_+(0) \leq m - N\kappa_r = CRG_N(0) + rNK^*/g_N(0)$. $\square$

The slope at a zero reserve

The continuity bound (27) can be sharpened to an exact slope.

Proposition S.1 (Small reserves). *Let $q^* := F(\hat{v})$ be the probability of the pooling region in Theorem 1. Then $\log(1/q^*) < N$ and*

$$\lim_{r \downarrow 0} \frac{CRG_N(r) - CRG_N(0)}{r} = \frac{Nq^*}{N - \log(1/q^*)}.$$

A small reserve raises revenue through two channels. Buyers in the pooling region now pay r instead of zero, which adds rq^* to revenue directly. This lowers the rent, by $q^*/(N - \log(1/q^*))$ per unit of reserve, and a lower rent raises the bidding rule above the pool. By (11), a unit reduction in the rent raises the bid at quantile q by

$1/q$, which adds $\log(1/q^*)$ to revenue, and the two channels together give the slope. Opting out has no first-order effect. Withdrawing the lowest s quantiles from the market reduces revenue by an amount proportional to rs , but it also destroys the value $\Phi_\mu(s)$, which lowers the rent and so raises revenue by an amount proportional to $\Phi_\mu(s)$. Since $\Phi_\mu(s)$ is of order s^2 , the best withdrawal involves a share of order r and changes revenue only by an amount of order r^2 . For uniform values and $N = 2$, where $q^* \approx 0.673$ by Example 2, the slope is about 0.84.

Proof of Proposition S.1. Write $L := \log(1/q^*)$ and keep the notation of the continuity proof. Under Assumption 1 the pooling quantile $q_K = \Phi_\mu^{-1}(K)$ is continuous in K . For $0 \leq K \leq K^*$ and every q ,

$$(K^* - K) \mathbb{1}\{\Phi_\mu(q) > K^*\} \leq (\Phi_\mu(q) - K)^+ - (\Phi_\mu(q) - K^*)^+ \leq (K^* - K) \mathbb{1}\{\Phi_\mu(q) > K\},$$

and integrating against dq/q gives

$$L(K^* - K) \leq \mathcal{R}_\mu(K) - \mathcal{R}_\mu(K^*) \leq \log(1/q_K)(K^* - K). \quad (28)$$

Since $g_N(K) = N(K^* - K) - [\mathcal{R}_\mu(K) - \mathcal{R}_\mu(K^*)]$, this is equivalent to

$$(N - \log(1/q_K))(K^* - K) \leq g_N(K) \leq (N - L)(K^* - K). \quad (29)$$

At $K = 0$ the right inequality gives $(N - L)K^* \geq g_N(0) > 0$, so $L < N$.

Upper bound. Fix $t > q^*/(N - L)$ and let $\kappa := K^* - tr$. Since $\max\{r, x\} \leq x^+ + r \mathbb{1}\{x \leq r\}$, and $(\Phi_\mu(q) - \kappa)/q \leq r$ exactly for $q \leq q_r(\kappa)$, we have $h_0(\kappa) \geq g_N(\kappa) - r q_r(\kappa)$. Moreover $q_r(\kappa) \leq q_{\kappa+r}$, because $\Phi_\mu(q) - rq \leq \kappa$ implies $\Phi_\mu(q) \leq \kappa + r$. By (29),

$$h_0(\kappa) \geq r \left[t(N - \log(1/q_\kappa)) - q_{\kappa+r} \right],$$

and the bracket converges to $t(N - L) - q^* > 0$ as $r \downarrow 0$. Hence $\kappa_+(0) \geq K^* - tr$ for small r , and the argument for the upper bound in (27) gives $CRG_N(r) \leq \mathcal{F}(0) = m - N\kappa_+(0) \leq CRG_N(0) + Ntr$. Letting $t \downarrow q^*/(N - L)$, the limit superior of $(CRG_N(r) - CRG_N(0))/r$ is at most $Nq^*/(N - L)$.

Lower bound. By Proposition 5, $CRG_N(r)$ is the infimum of the revenue R over identical play BCCE with reserve r , and by (27) the infimum may be taken over those with $R \leq CRG_N(0) + Cr$, where $C := NK^*/g_N(0) + 1$. Fix such a BCCE, with opt-out share s and rent K , and write $x(q) := (\Phi_\mu(q) - K)/q$ and $I := \int_0^s x(q)^+ dq$.

Step 1 of the exactness proof in Appendix C gives $R \geq \int_s^1 \max\{r, x(q)\} dq$, $NK \leq m - \Phi_\mu(s) - R$, and $\Phi_\mu(s) - rs \leq K$. Since $\max\{r, x\} \geq x^+ + r \mathbb{1}\{x \leq 0\}$, and $x \leq 0$ exactly on $[0, q_K]$, the first inequality gives

$$R \geq \mathcal{R}_\mu(K) - I + r(q_K - s)^+. \quad (30)$$

With the second, $g_N(K) \geq \Phi_\mu(s) - I + r(q_K - s)^+ \geq 0$, since $x^+ \leq F^{-1}$ gives $I \leq \Phi_\mu(s)$. Hence $K \leq K^*$ by Lemma 7, and (29) gives $(N - L)(K^* - K) \geq \Phi_\mu(s) - I + r(q_K - s)^+$. Combining this with (30) and the left inequality in (28),

$$R - CRG_N(0) \geq L(K^* - K) - I + r(q_K - s)^+ \geq \frac{L\Phi_\mu(s) - NI + Nr(q_K - s)^+}{N - L}.$$

If $s \leq q_K$, then $I = 0$. If $s > q_K$, the inequality $\Phi_\mu(s) - rs \leq K$ gives $x(s) \leq r$, so $x \leq r$ on $(q_K, s]$ and $I \leq r(s - q_K)$. In both cases

$$R - CRG_N(0) \geq \frac{Nr q_K + L\Phi_\mu(s) - Nrs}{N - L} \geq \frac{Nr}{N - L} \left[q_K - F(Nr/L) \right],$$

where the second inequality minimizes the convex function $s \mapsto L\Phi_\mu(s) - Nrs$, whose minimum is attained at $s = F(Nr/L)$ and is at least $-NrF(Nr/L)$. Finally, (30) and $I \leq r$ give $\mathcal{R}_\mu(K) \leq R + r \leq CRG_N(0) + (C + 1)r$, so the left inequality in (28) gives $K \geq K^* - (C + 1)r/L$ and hence $q_K \geq q_{K^* - (C+1)r/L}$. These bounds do not depend on the BCCE, so

$$\frac{CRG_N(r) - CRG_N(0)}{r} \geq \frac{N}{N - L} \left[q_{K^* - (C+1)r/L} - F(Nr/L) \right],$$

and the right side converges to $Nq^*/(N - L)$ as $r \downarrow 0$. $\square$

S.3 Atomic Priors, Convex Order, and the Binary Bound

Proof of Theorem 6

Appendix B is written for an arbitrary non-degenerate prior, with F^{-1} the quantile function and q_K defined by the supremum, and it invokes the density in Assumption 1

only to write the attaining bid as $c = \psi(F(v))$. Proposition 1, Lemma 1, and Lemma 2 do not use the density either. Hence (10) has a unique solution $K^* \in (0, m)$ by Lemma 7, every BCCE has revenue at least $m - NK^*$ by Step 1 of Appendix B and Proposition 1, and Step 2 shows that the identical play distribution $v = F^{-1}(U)$, $c = \psi(U)$ attains this bound. At an atom of μ , U is uniform on the corresponding quantile interval, so this coupling randomizes the common bid within the atom. $\square$

Proofs for Section 9

Proof of Proposition 8. Suppose $\mu \preceq_{\text{cx}} \nu$. By the quantile characterization of the convex order (Shaked and Shanthikumar, 2007, Theorem 3.A.5), the two distributions share the mean m and satisfy $\Phi_\nu(q) \leq \Phi_\mu(q)$ for all q . Hence $\mathcal{R}_\nu \leq \mathcal{R}_\mu$ pointwise by (9), so

$$g_N^\nu(K) = m - NK - \mathcal{R}_\nu(K) \geq m - NK - \mathcal{R}_\mu(K) = g_N^\mu(K) \quad \text{for all } K.$$

By Lemma 7 each is concave with a unique zero, crossing from above, so a pointwise larger function has a weakly larger zero: $K_\nu^* \geq K_\mu^*$. Therefore $\text{CRG}(\nu) = m - NK_\nu^* \leq m - NK_\mu^* = \text{CRG}(\mu)$.

For the second claim, among distributions supported on $[0, \bar{v}]$ with mean E , the binary distribution $\underline{\mu}$ on $\{0, \bar{v}\}$ is the largest in the convex order: for any such μ and any convex ϕ , $\int \phi d\mu \leq \int \phi d\underline{\mu}$, because on $[0, \bar{v}]$ a convex function lies below the chord joining $(0, \phi(0))$ and $(\bar{v}, \phi(\bar{v}))$. Hence $\text{CRG}(\underline{\mu}) \leq \text{CRG}(\mu)$ for every such μ . $\square$

Proof of Corollary 1. For $\underline{\mu}$ we have $F^{-1}(t) = 0$ for $t < p_0 := 1 - E/\bar{v}$ and $F^{-1}(t) = \bar{v}$ for $t > p_0$, so $\Phi(q) = \bar{v}(q - p_0)^+$ and $q_K = p_0 + K/\bar{v}$. Then

$$\mathcal{R}(K) = \int_{q_K}^1 \frac{\bar{v}(q - p_0) - K}{q} dq = \bar{v} \int_{q_K}^1 \frac{q - q_K}{q} dq = \bar{v}[(1 - q_K) + q_K \log q_K].$$

Substituting $K = \bar{v}(q_K - p_0)$ and $E = \bar{v}(1 - p_0)$ into (10) and simplifying yields $(N - 1)(p_0 - q_K) = q_K \log q_K$, with $\text{CRG} = E - N\bar{v}(q_K - p_0)$. Uniqueness of the root is Lemma 7. As $N \rightarrow \infty$ we have $K^* \rightarrow 0$ and $q_K \rightarrow p_0$, so

$$\text{CRG} \longrightarrow \bar{v}[(1 - p_0) + p_0 \log p_0] = E + (\bar{v} - E) \log \left(1 - \frac{E}{\bar{v}}\right). \quad \square$$

Remark 2. Because the binary prior is atomic, the density-form system of (5) and (6) does not apply to it and the worst-case common bid must randomize. This is why Theorem 6 draws the bid through the uniform variable U rather than as a function of the value. In the limiting case the common bid at $v = \bar{v}$ has CDF $F_c(\delta) = \frac{\bar{v}-E}{E} \cdot \frac{\delta}{\bar{v}-\delta}$ on the support $[0, E]$.